



Master's Degree Course in Robotics and Automation Engineering
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Payload Generalization of Learned Trajectories

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Part I

Rigid Manipulators

Chapter 1

Introduction

Given a serial manipulator, we are interested in studying how an object located at the end-effector affects the torque at the joints necessary to track a desired trajectory. To do this, first we look for any characteristics of the object on which the torque at the joints of the manipulator depends, and we will use them appropriately with a regressor to be able to generate the torque at the joints necessary to ensure the tracking of the desired trajectory with any new object.

Chapter 2

Model and Regressor

2.1 Model Description

The dynamic model of a serial robot with N joints is

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau, \quad (2.1)$$

where $M(q)$ is the Inertia Matrix, $C(q, \dot{q})$ collects Coriolis and Centrifugal forces, $G(q)$ is the Gravity force field and q are the joint angles, with their time derivatives $\dot{q}, \ddot{q}$. We can rewrite (2.1) in the new form

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Y(q, \dot{q}, \ddot{q})\pi, \quad (2.2)$$

where $Y(q, \dot{q}, \ddot{q})$ is a Regressor and π is a column vector containing the Dynamic Parameters.

Now we will assume that we can divide the vector π in two parts π_m containing the dynamic parameters of the manipulator and π_o containing the parameters of an object located at the End-effector.

With these assumptions, the right part of (2.2) can be written in a matrix form

$$Y(q, \dot{q}, \ddot{q})\pi = \begin{bmatrix} Y_m & Y_o \end{bmatrix} \begin{bmatrix} \pi_m \\ \pi_o \end{bmatrix} = \tau_m + \tau_o = \tau. \quad (2.3)$$

Writing (2.2) in the form in (2.3) allows us to see the torque as the sum of two contributions: the first τ_m relating to the torque needed to move the manipulator without an object at the end effector. The second τ_o relating to the torque needed to compensate the force due to the object at the End-Effector.

2.2 Regressor Description

The coefficients of the Regressor $Y(q, \dot{q}, \ddot{q})$ can be obtained as a linearization of (2.1)

$$Y_{ij} = \frac{1}{k!} \left. \frac{\partial \tau_i}{\partial \pi_j^k} \right|_{\pi=0} \quad (2.4)$$

2.2.1 Dynamic Model

Assuming that the dynamic model of a manipulator is linear with respect to some parameters named Dynamic Parameters[1], our purpose is to determine them.

Let's consider a two-link planar arm where:

- q_1, q_2 are the joint angles
- l_1, l_2 are the distances of the centres of mass of the two links from the respective joint axes.
- a_1, a_2 are the lengths of the two links.
- m_{l_1}, m_{l_2} are the masses of the two links.
- m_{m_1}, m_{m_2} are the masses of the motors of the two joint.
- I_{m_1}, I_{m_2} are the moments of inertia of the two motors.
- I_{l_1}, I_{l_2} are the moments of inertia relative to the centres of mass of the two links.
- k_{r_1}, k_{r_2} are the gear reduction ratio of the two motors.
- m_e and I_e be the mass and the moment of inertia of the object at the End-Effector.

we obtain the following equations of motion where τ_1 and τ_2 are the torques applied to the joints

$$\begin{aligned}
\tau_1 = & (I_{l_1} + I_{l_2} + I_e + I_{m_2} + I_{m_1}k_{r_1}^2 + a_1^2m_e + a_2^2m_e + a_1^2m_{l_2} + a_1^2m_{m_2} \\
& + l_1^2m_{l_1} + l_2^2m_{l_2} + 2a_1a_2m_e \cos(q_2) + 2a_1l_2m_{l_2} \cos(q_2))\ddot{q}_1 + (m_e a_2^2 \\
& + a_1m_e \cos(q_2)a_2 + m_{l_2}l_2^2 + a_1m_{l_2} \cos(q_2)l_2 + I_{l_2} + I_e + I_{m_2}k_{r_2})\ddot{q}_2 \\
& + (a_1m_e \cos(q_1) + a_1m_{l_2} \cos(q_1) + a_1m_{m_2} \cos(q_1) + l_1m_{l_1} \cos(q_1) \\
& + a_2m_e \cos(q_1 + q_2) + l_2m_{l_2} \cos(q_1 + q_2))g - a_1a_2\dot{q}_2^2m_e \sin(q_2) \\
& - a_1\dot{q}_2^2l_2m_{l_2} \sin(q_2) - 2a_1a_2\dot{q}_1\dot{q}_2m_e \sin(q_2) - 2a_1\dot{q}_1\dot{q}_2l_2m_{l_2} \sin(q_2). \\
\tau_2 = & (m_e a_2^2 + a_1m_e \cos(q_2)a_2 + m_{l_2}l_2^2 + a_1m_{l_2} \cos(q_2)l_2 + I_{l_2} + I_e + I_{m_2}k_{r_2})\ddot{q}_1 \\
& + (m_e a_2^2 + I_{m_2}k_{r_2}^2 + m_{l_2}l_2^2 + I_{l_2} + I_e)\ddot{q}_2 + (a_2m_e \cos(q_1 + q_2) \\
& + l_2m_{l_2} \cos(q_1 + q_2))g + a_1a_2\dot{q}_1^2m_e \sin(q_2) + a_1\dot{q}_1^2l_2m_{l_2} \sin(q_2).
\end{aligned} \tag{2.5}$$

2.2.2 Dynamic Parameters

At this point, we are interested in writing (2.5) according to (2.2). For the simple case of a two-link planar arm it's possible to observe (2.5) and to choose accordingly the parameters, arrange them so as to have the form in (2.3). We obtain the following parameters vector

$$\pi = [\pi_1 \quad \pi_2 \quad \pi_3 \quad \pi_4 \quad \dots \quad \pi_{10} \quad \pi_{11} \quad \pi_{12} \quad \pi_{13}]^T \tag{2.6}$$

where :

$$\begin{aligned}
\pi_1 = m_{l_1}, \quad \pi_2 = m_{m_2}, \quad \pi_3 = m_{l_1}l_1, \quad \pi_4 = I_{l_1}, \quad \pi_5 = I_{m_2}, \quad \pi_6 = m_{l_1}l_1^2, \quad \pi_7 = I_{m_1}, \\
\pi_8 = m_{l_2}, \quad \pi_9 = m_{l_2}l_2, \quad \pi_{10} = I_{l_2}, \quad \pi_{11} = m_{l_2}l_2^2, \quad \pi_{12} = m_e, \quad \pi_{13} = I_e.
\end{aligned}$$

So the final regressor is a matrix $Y_{n \times p}$ which for our case is $Y_{2 \times 13}$:

$$\begin{aligned}
Y_{1,1} &= 0, \\
Y_{2,1} &= 0, \\
Y_{1,2} &= \ddot{q}_1 a_1^2 + g \cos(q_1) a_1, \\
Y_{2,2} &= 0, \\
Y_{1,3} &= g \cos(q_1), \\
Y_{2,3} &= 0, \\
Y_{1,4} &= \ddot{q}_1, \\
Y_{2,4} &= 0, \\
Y_{1,5} &= \ddot{q}_1 + \ddot{q}_2 k_{r_2}, \\
Y_{2,5} &= k_{r_2} \ddot{q}_1 + \dot{q}_2^2 k_{r_2}^2, \\
Y_{1,6} &= \ddot{q}_1, \\
Y_{2,6} &= 0, \\
Y_{1,7} &= \ddot{q}_1 k_{r_1}^2, \\
Y_{2,7} &= 0, \\
Y_{1,8} &= \ddot{q}_1 a_1^2 + g \cos(q_1) a_1, \\
Y_{2,8} &= 0, \\
Y_{1,9} &= -a_1 \sin(q_2) \dot{q}_2^2 - 2a_1 \dot{q}_1 \sin(q_2) \dot{q}_2 + g \cos(q_1 + q_2) + 2a_1 \ddot{q}_1 \cos(q_2) + \\
& a_1 \ddot{q}_2 \cos(q_2),
\end{aligned}$$

$$\begin{aligned}
Y_{2,9} &= a_1 \sin(q_2) \dot{q}_1^2 + g \cos(q_1 + q_2) + a_1 \ddot{q}_1 \cos(q_2), \\
Y_{1,10} &= \ddot{q}_1 + \ddot{q}_2, \\
Y_{2,10} &= \ddot{q}_1 + \ddot{q}_2, \\
Y_{1,11} &= \ddot{q}_1 + \ddot{q}_2, \\
Y_{2,11} &= \ddot{q}_1 + \ddot{q}_2, \\
Y_{1,12} &= (a_1^2 + 2a_1 a_2 \cos(q_2) + a_2^2) \ddot{q}_1 + (a_2^2 + a_1 a_2 \cos(q_2)) \ddot{q}_2 + (a_2 \cos(q_1 + q_2) + \\
&\quad a_1 \cos(q_1)) g - a_1 a_2 \sin(q_2) \dot{q}_2^2 - 2a_1 a_2 \dot{q}_1 \dot{q}_2 \sin(q_2), \\
Y_{2,12} &= (a_2^2 + a_1 a_2 \cos(q_2)) \ddot{q}_1 + a_2^2 \ddot{q}_2 + a_2 \cos(q_1 + q_2) g + a_1 a_2 \dot{q}_1^2 \sin(q_2), \\
Y_{1,13} &= \ddot{q}_1 + \ddot{q}_2, \\
Y_{2,13} &= \ddot{q}_1 + \ddot{q}_2.
\end{aligned}$$

As we can see the regressor can be separate in two parts as in (2.3)

$$\begin{aligned}
Y_m &= \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & \dots & Y_{19} & Y_{1,10} & Y_{1,11} \\ Y_{21} & Y_{22} & Y_{23} & \dots & Y_{29} & Y_{2,10} & Y_{2,11} \end{bmatrix}, \\
\pi_m &= [\pi_1 \quad \pi_2 \quad \pi_3 \quad \dots \quad \pi_9 \quad \pi_{10} \quad \pi_{11}]^T.
\end{aligned}$$

$$Y_o = \begin{bmatrix} Y_{1,12} & Y_{1,13} \\ Y_{2,12} & Y_{2,13} \end{bmatrix}, \quad \pi_o = [\pi_{12} \quad \pi_{13}]^T \quad (2.7)$$

For our purpose, as we will see in the next section, we are only interested in $\pi_o = [m_e \ I_e]^T$ which contains the parameters of the object at the End-Effector.

Chapter 3

Object-Manipulator Torque Identification : Method 1

3.1 Description

Starting from (2.3), let's consider an object i then we can write

$${}^i\tau = \tau_m + {}^i\tau_o = \tau_m + Y_o {}^i\pi_o \quad (3.1)$$

we can try to write (3.1) as the product of two terms

$${}^i\tau = \bar{Y}_o {}^i\bar{\pi}_o, \quad (3.2)$$

where $\bar{Y}_o = [Y_m \pi_m \quad Y_o]$ and ${}^i\bar{\pi}_o = \begin{bmatrix} 1 \\ {}^i\pi_o \end{bmatrix}$. From (3.2) we have an expression of the torque function of the vector ${}^i\bar{\pi}_o$. At this level, we are interested in obtaining $\bar{Y}_o$ since ${}^i\bar{\pi}_o$ is known. One way to do this is to acquire the torque ${}^i\tau$ needed to track the desired trajectory for different objects and write them in a matrix T , such that $T = \begin{bmatrix} {}^0\tau & {}^1\tau & {}^2\tau & \dots & {}^r\tau \end{bmatrix}$, then we do the same for the vector ${}^i\bar{\pi}_o$ such that $\Pi = \begin{bmatrix} {}^0\bar{\pi}_o & {}^1\bar{\pi}_o & {}^2\bar{\pi}_o & \dots & {}^r\bar{\pi}_o \end{bmatrix}$, at this point we get the matrix form of (3.2)

$$T = \bar{Y}_o \Pi, \quad (3.3)$$

where by inverting Π we can get

$$\bar{Y}_o = T \Pi^{-1}. \quad (3.4)$$

Finally, from (2.1), (2.2), (2.3), (3.2), (3.3) and (3.4) the Object-Manipulator torque is given by

$$\tau_A = \bar{Y}_o \bar{\pi}. \quad (3.5)$$

Parameters	Obj 0	Obj 1	Obj 2	Obj 3	Obj 4	Obj 5
mass (kg)	0	5	7	23	33	50
Inertia ($kg.m^2$)	0	1	1.7	3.5	4.1	5.32

Table 3.1: Object Parameters

3.2 Evaluation and Results

3.2.1 Desired Torque Computation/Acquisition

Consist in computing the desired torque for each mass according to (2.1) where $q = q_d$, $\dot{q} = \dot{q}_d$ and $\ddot{q} = \ddot{q}_d$.

$$\tau_d = M(q_d)\ddot{q}_d + C(q_d, \dot{q}_d)\dot{q}_d + G(q_d). \quad (3.6)$$

3.2.2 Evaluation

For the Evaluation of the proposed method, we will first evaluate the torque estimation error

$$e_t = \tau_d - \tau_A \quad (3.7)$$

and secondly the tracking error of the joint angles and their time derivatives

$$e_q = q_d - q_A ; \quad \dot{e}_q = \dot{q}_d - \dot{q}_A ; \quad \ddot{e}_q = \ddot{q}_d - \ddot{q}_A. \quad (3.8)$$

where q_A are joint angles due to the torque τ_A .

3.2.3 Results

The following data are used for the simulation:

$$\begin{aligned} a_1 &= a_2 = 1m , \quad l_1 = l_2 = 0.5m \\ m_{l_1} &= m_{l_2} = 50kg , \quad m_{m_1} = m_{m_2} = 5kg \\ I_{l_1} &= I_{l_2} = 10kg.m^2 , \quad I_{m_1} = I_{m_2} = 0.01kg.m^2 \\ k_{r_1} &= k_{r_2} = 100. \end{aligned}$$

The desired joint angles are:

$$q_d = [\sin(t) , 0.5\sin(t)]^T , \text{ with } q(0) = [0, 0]^T, \quad \dot{q}(0) = [1, 0.5]^T.$$

The proof object at the End-effector are in table 3.1.

Since for our case we have $\Pi_{3 \times 3}$, this means that we will use object 0 and two other objects for the algorithm and can use one or more objects for the falsification phase.

$$T = \begin{bmatrix} 0 & \tau & 1 & \tau & 2 & \tau \end{bmatrix} \text{ and } \Pi = \begin{bmatrix} 1 & 1 & 1 \\ m_0 & m_1 & m_2 \\ I_0 & I_1 & I_2 \end{bmatrix}. \text{ We have chosen object 1 and 2}$$

for the algorithm and object 5 for the falsification and we get results as shown in figure 3.1.

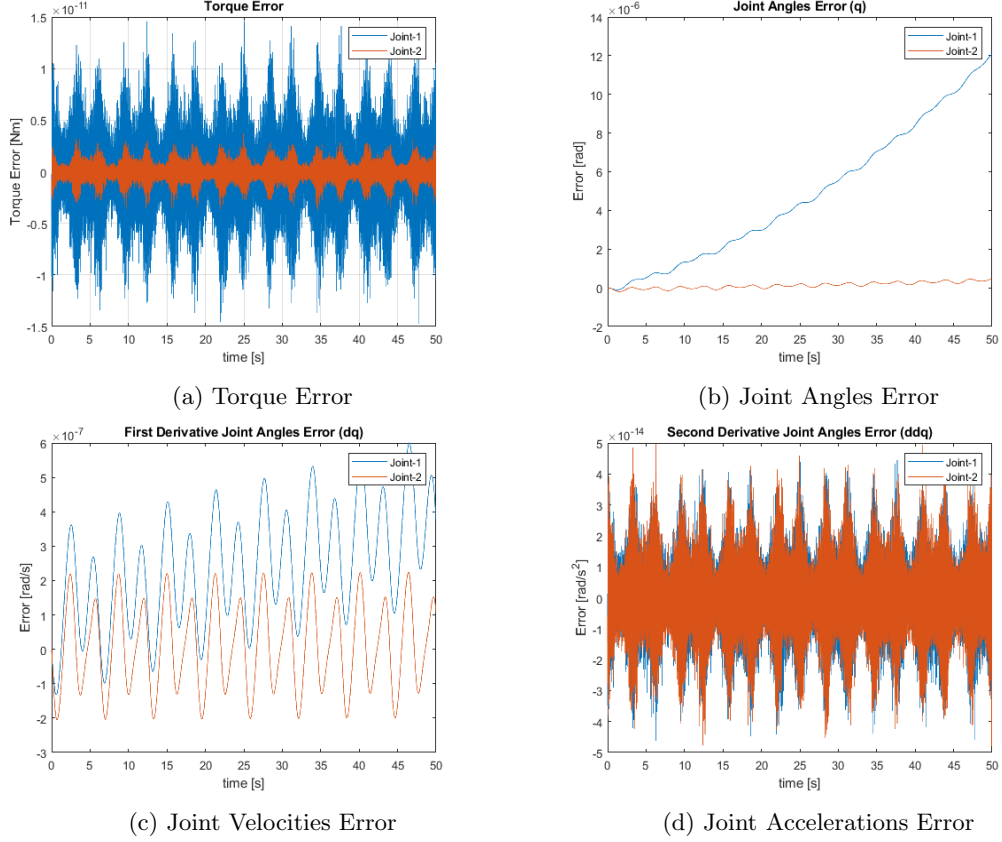


Figure 3.1: Tracking Performance on Rigid manipulator with method 1 using Object 5.

Paying attention to figure 3.1b, we can see that the joint angles diverge. This is due to the fact that the proposed method uses a feed-forward torque as in equation (3.6) and therefore we have a numerical error propagation that leads to the observed divergence. After all, the proposed method is quite good since the error of all significant variables (see equations (3.7) and (3.8)) is very small, on the order of 10^{-6} rad for the joint angles error.

Looking carefully equation (2.3) instead of Identifying τ , one can try to identify only τ_o and sum it with τ_m which can be seen as the desired torque for empty End-Effector manipulator (notice that $\tau_m = {}^0\tau$). This second method that we will develop below can be named “ *Object Torque Identification*”.

Chapter 4

Object Torque Identification : Method 2

4.1 Description

We can see from (2.3) that

$$\tau_o = \tau - \tau_m \quad (4.1)$$

and so for each object we have (from equation (3.1))

$${}^i\tau_o = Y_o {}^i\pi_o \quad (4.2)$$

as before, we are interested in obtaining Y_o , so we build the matrix T_o as $T_o = [{}^1\tau_o \quad {}^2\tau_o \quad \dots \quad {}^r\tau_o]$, then we do the same for the parameters vector such that $\Pi_o = [{}^1\pi_o \quad {}^2\pi_o \quad \dots \quad {}^r\pi_o]$, at this point we get the matrix form of (4.2)

$$T_o = Y_o \Pi_o \quad (4.3)$$

where by inverting Π_o we can get

$$Y_o = T_o \Pi_o^{-1} \quad (4.4)$$

Finally, from (4.2) , (4.3) and (4.4) the Object torque is given by

$$\bar{\tau}_o = Y_o \pi_o \quad (4.5)$$

and, from (2.3) and (4.5) the Object-Manipulator torque is given by

$$\tau_A = \tau_m + \bar{\tau}_o \quad (4.6)$$

4.2 Evaluation and Results

We will use the same Criteria as in section 3.2.2 using equations (3.7) and (3.8) for the evaluation. For the results, the same data in table 3.1 are used for the simulation.

Here, $T_o = \begin{bmatrix} {}^1\tau_o & {}^2\tau_o \end{bmatrix}$ and $\Pi_o = \begin{bmatrix} m_1 & m_2 \\ I_1 & I_2 \end{bmatrix}$. As before, object 5 for the falsification, the results are in the following figures.

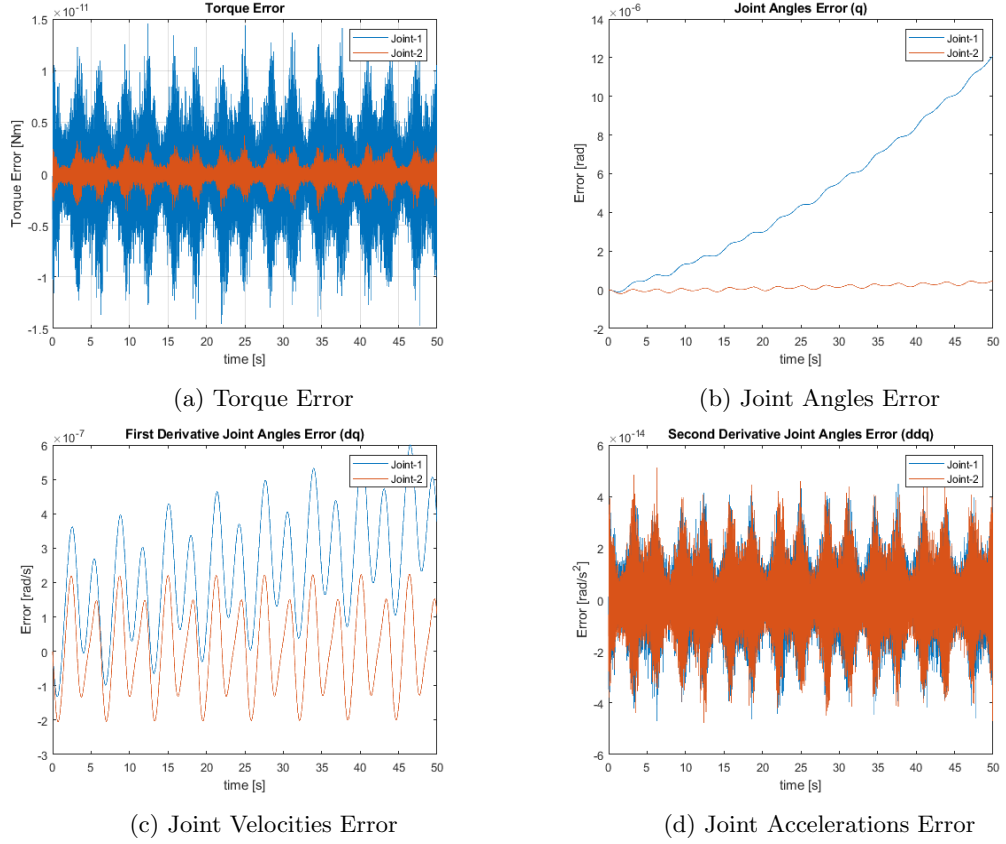


Figure 4.1: Tracking Performance on Rigid manipulator with method 2 using Object 5.

As we can see from figure 4.1, this approach also provides satisfactory results that are almost equal or let's say equal to the first approach.

In the next chapter, let's see what happen when we characterise the object as a link.

Chapter 5

Manipulator with Object as a Link

5.1 Introduction

In chapter 2, we have seen that a rigid manipulator can be described by equation (2.1). Since we are also interested in the object located at the end effector, in this chapter we will try to model the object as a link and then we will apply the two methods seen in chapter 3 and 4 on this manipulator model. Basically the structure of the manipulator is the same and only the perception of the object will change to try to better capture reality.

5.2 Dynamic Model

We can associate the following dynamic parameters to a link :

- l is the distance of the centre of mass of a link from the respective joint axe.
- a is the length of a link.
- m_l is the mass of a link.
- m_m is the mass of the motor of a joint.
- I_l is the moment of inertia of the centre of mass of a link.
- I_m is the moment of inertia of a motor.
- m_e is the mass of the end-effector at a link.
- I_e is the moment of inertia of the end-effector of a link.
- k_r is the gear reduction ratio of a motor.

Parameters	Link 1	Link 2	Object
$l(m)$	l_1	l_2	l_o
$a(m)$	a_1	a_2	a_o
m_l (kg)	m_{l_1}	m_{l_2}	m_o
m_m (kg)	m_{m_1}	m_{m_2}	0
$I_l(kg.m^2)$	I_{l_1}	I_{l_2}	I_o
$I_m(kg.m^2)$	I_{m_1}	I_{m_2}	0
$m_e(kg)$	0	m_{e_2}	0
$I_e(kg.m^2)$	0	I_{e_2}	0
k_r	k_{r_1}	k_{r_2}	0

Table 5.1: Simple Dynamic Parameters

So to the object located at the final end effector, we will attribute each of these parameters.

For a two-link planar arm, the simple dynamic parameters are resumed in table 5.1

The Object Parameters are expressed with respect to the end-effector.

We obtain the following equations of motion where τ_1 and τ_2 are the torques applied to the joints

$$\begin{aligned}
\tau_1 = & \ddot{q}_1(I_{l_1} + I_{l_2} + I_{e_2} + I_{m_2} + I_o + I_{m_1}k_{r_1}^2 + a_1^2m_{e_2} + a_2^2m_{e_2} + a_1^2m_{l_2} \\
& + a_1^2m_{m_2} + a_1^2m_o + a_2^2m_o + l_1^2m_{l_1} + l_2^2m_{l_2} + l_o^2m_o + 2a_2l_o m_o + 2a_1a_2m_{e_2} \cos(q_2) \\
& + 2a_1a_2m_o \cos(q_2) + 2a_1l_2m_{l_2} \cos(q_2) + 2a_1l_o m_o \cos(q_2)) \\
& + \ddot{q}_2(I_{l_2} + I_{e_2} + I_o + I_{m_2}k_{r_2}^2 + a_2^2m_{e_2} + a_2^2m_o + l_2^2m_{l_2} + l_o^2m_o + 2a_2l_o m_o \\
& + a_1a_2m_{e_2} \cos(q_2) + a_1a_2m_o \cos(q_2) + a_1l_2m_{l_2} \cos(q_2) + a_1l_o m_o \cos(q_2)) \\
& + gm_o(\cos(q_1 + q_2)(a_2 + l_o) + a_1 \cos(q_1)) + gm_{e_2}(a_2 \cos(q_1 + q_2) + a_1 \cos(q_1)) \\
& + gm_{l_2}(l_2 \cos(q_1 + q_2) + a_1 \cos(q_1)) + a_1gm_{m_2} \cos(q_1) + gl_1m_{l_1} \cos(q_1) \\
& - a_1\dot{q}_2 \sin(q_2)(\dot{q}_1 + \dot{q}_2)(a_2m_{e_2} + a_2m_o + l_2m_{l_2} + l_o m_o) \\
& - a_1\dot{q}_1\dot{q}_2 \sin(q_2)(a_2m_{e_2} + a_2m_o + l_2m_{l_2} + l_o m_o) \\
& \tau_2 = a_1 \sin(q_2)(a_2m_{e_2} + a_2m_o + l_2m_{l_2} + l_o m_o)\dot{q}_1^2 \\
& + \ddot{q}_2(I_{l_2} + I_{e_2} + I_o + I_{m_2}k_{r_2}^2 + a_2^2m_{e_2} + a_2^2m_o + l_2^2m_{l_2} + l_o^2m_o + 2a_2l_o m_o) \\
& + \ddot{q}_1(I_{l_2} + I_{e_2} + I_o + I_{m_2}k_{r_2}^2 + a_2^2m_{e_2} + a_2^2m_o + l_2^2m_{l_2} + l_o^2m_o + 2a_2l_o m_o \\
& + a_1a_2m_{e_2} \cos(q_2) + a_1a_2m_o \cos(q_2) + a_1l_2m_{l_2} \cos(q_2) + a_1l_o m_o \cos(q_2)) \\
& + g \cos(q_1 + q_2)(a_2m_{e_2} + a_2m_o + l_2m_{l_2} + l_o m_o)
\end{aligned} \tag{5.1}$$

5.3 Dynamic Parameters

Now we can compute the regressor coefficients using equation (2.4) (see section 2.2.2).

We obtain a regressor $Y_{n \times p} = Y_{2 \times 18}$ and the dynamic parameters vector π is vector column of dimension p .

$$\pi_1 = [m_{m_1} \quad I_{L_1} \quad I_{m_1} \quad m_{l_1} \quad m_{l_1} l_1 \quad m_{l_1} l_1 l_1]^T$$

$$\pi_2 = [m_{m_2} \quad I_{L_2} \quad I_{m_2} \quad m_{l_2} \quad m_{l_2} l_2 \quad m_{l_2} l_2 l_2]^T$$

$$\pi_e = [I_{e_2} \quad m_{e_2}]^T$$

Which up to here is equal to what we did in the chapter 2. Now for the object we have

$$\pi_o = [I_o \quad m_o \quad m_o l_o \quad m_o l_o l_o]^T$$

And finally

$$\pi = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_e \\ \pi_o \end{bmatrix}$$

If we look carefully at the obtained regressor, we realize that some dynamic parameters have the same coefficient in the regressor and therefore we can reduce the size of the dynamic parameters vector by putting them together. For each link we obtain the following compacted dynamic parameters [5]:

- I_{m_i} : moment of inertia of the motor associated to link i .
- $m_{m_i} + m_{l_i}$: motor and link mass i .
- $m_{l_i} l_i$: first moment of inertia with respect to the origin of frame i .
- $I_{l_i} + m_{l_i} l_i^2 + I_{e_i}$: complete inertia tensor of link i
- m_{e_i} : mass of End-effector i

Observation 1: Only the last link has an end-effector.

Observation 2: If a compacted dynamic parameter is null we don't consider it in π .

Therefore, for a manipulator with n joints, the vector of the dynamic parameters is given in equation (5.3)

$$\pi_{l_i} = [I_{m_i} \quad m_{m_i} + m_{l_i} \quad m_{l_i} l_i \quad I_{l_i} + m_{l_i} l_i^2 + I_{e_i} \quad m_{e_i}]^T \quad i = 1, \dots, n \quad (5.2)$$

$$\pi_m = \begin{bmatrix} \pi_{l_1} \\ \vdots \\ \pi_{l_n} \end{bmatrix} \quad (5.3)$$

We can finally write the vector of the dynamic parameters of the manipulator as :

Parameters	Obj 0	Obj 1	Obj 2	Obj 3	Obj 4	Obj 5
mass (kg)	0	5	7	23	33	50
Inertia ($kg.m^2$)	0	1	1.7	3.5	4.1	5.32
COM (m)	0	0.1	0.2	0.3	0.4	0.5

Table 5.2: Object Parameters

$$\begin{aligned}\pi_{l_1} &= [I_{m_1}, m_{m_1} + m_{l_1}, m_{l_1}l_1, I_{l_1} + m_{l_1}l_1^2]^T, \\ \pi_{l_2} &= [I_{m_2}, m_{m_2} + m_{l_2}, m_{l_2}l_2, I_{l_2} + m_{l_2}l_2^2 + I_{e_2}, m_{e_2}]^T, \\ \pi_m &= \begin{bmatrix} \pi_{l_1} \\ \pi_{l_2} \end{bmatrix}\end{aligned}$$

and the object dynamic parameters vector as

$$\pi_o = [m_o \quad m_o l_o \quad I_o + m_o l_o^2]^T \quad (5.4)$$

Finally reorder the regressor $Y_{2 \times 12}$ according to π_m and π_o .

The last expression of π_o in (5.4) shows us the compacted dynamic parameters of the object and now we can apply both methods seen in chapter 3 and 4.

5.4 Evaluation and Results

For the evaluation, We will use the same Criteria as in section (3.2.2) using (3.7) and (3.8). For the results, the data used for the simulation are in table 5.2.

For method 1:

$$T = \begin{bmatrix} 0\tau & 1\tau & 2\tau & 3\tau \end{bmatrix} \text{ and } \Pi = \begin{bmatrix} 1 & 1 & 1 & 1 \\ m_0 & m_1 & m_2 & m_3 \\ m_0 l_0 & m_1 l_1 & m_2 l_2 & m_3 l_3 \\ I_0 + m_0 l_0^2 & I_1 + m_1 l_1^2 & I_2 + m_2 l_2^2 & I_3 + m_3 l_3^2 \end{bmatrix}$$

For method 2:

$$T_o = \begin{bmatrix} 1\tau_o & 2\tau_o & 3\tau_o \end{bmatrix} \text{ and } \Pi_o = \begin{bmatrix} m_1 & m_2 & m_3 \\ m_1 l_1 & m_2 l_2 & m_3 l_3 \\ I_1 + m_1 l_1^2 & I_2 + m_2 l_2^2 & I_3 + m_3 l_3^2 \end{bmatrix}$$

As before, object 5 is used for the falsification. The results are shown in figures 5.1 and 5.2.

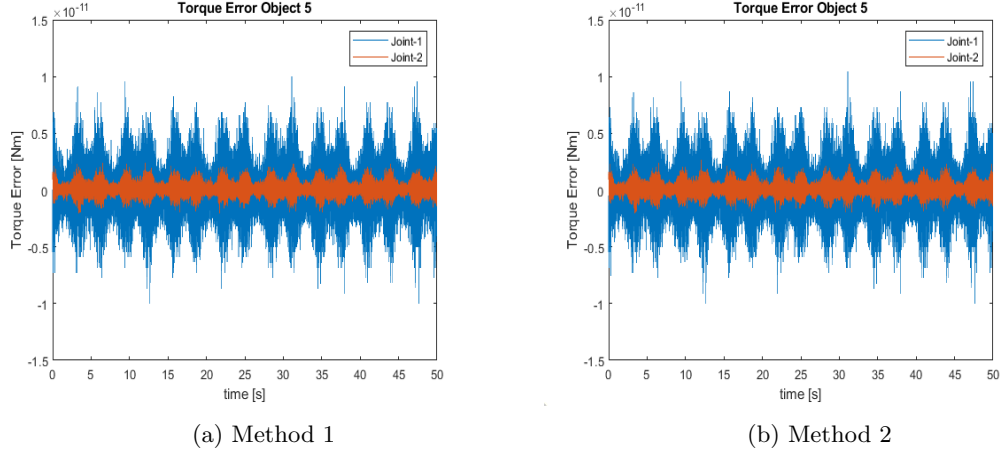
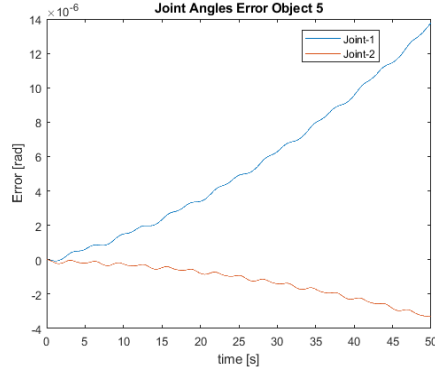


Figure 5.1: Torque Error

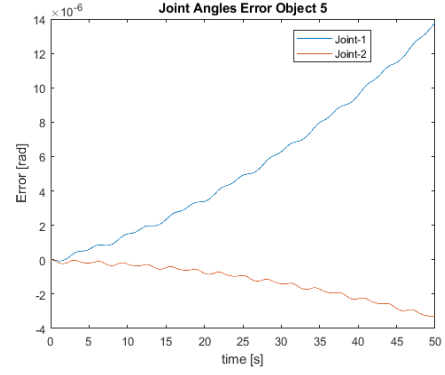
5.5 Conclusion

Also in this case, the results achieved are good as those in the previous chapters for tracking the desired trajectory $q_d(t)$, and this way of describing the manipulator and the object offers a better understanding of the dynamics of the manipulator.

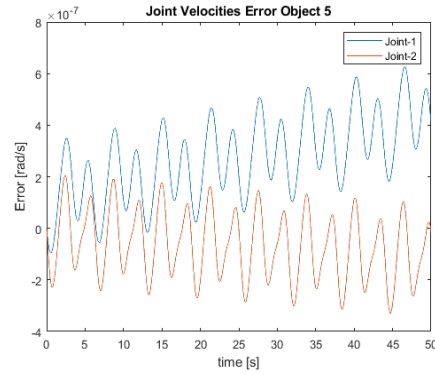
Since the rigid model in (2.1) is an approximation of real manipulators, where it is assumed that the joint angles and the angles of the motors are equal and therefore neglect the dynamics of the motors. At this point we are interested in what it would become if we extended the model to one that integrates the dynamics of the actuators.



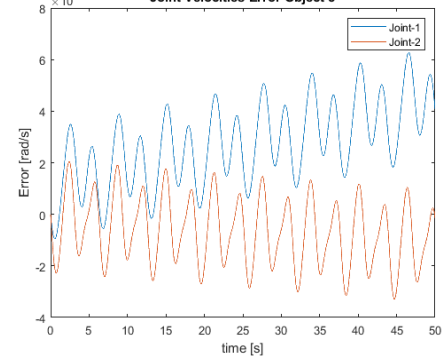
(a) Joint Angles Error Method 1



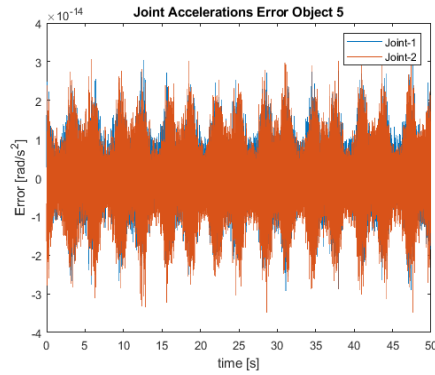
(b) Joint Angles Error Method 2



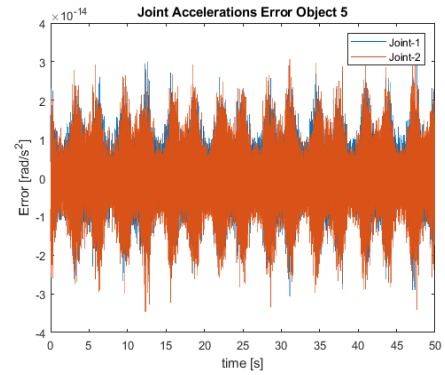
(c) Joint Velocities Error Method 1



(d) Joint Velocities Error Method 2 Joint Accelerations Error



(e) Joint Accelerations Error Method 1



(f) Joint Accelerations Error Method 2

Figure 5.2: Tracking Performance on Rigid manipulator with object described as a link using Object 5.

Part II

Flexible Manipulators

In the previous chapters, we have seen that a rigid manipulator can be described by equation (2.1) which is nonlinear in the variable q and we have seen how we can rewrite it in the form of equation (2.2) taking advantage of the linear dependence it has compared to what we have called dynamic parameters and above all we have since the torque applied to the joints could be calculated as the sum of two contributions (see equation (2.3)): the first, function of the dynamic parameters of the manipulator itself and the second function of parameters of the object that the manipulator will have to move which are mass and inertia. Thanks to the knowledge of these last two parameters, we managed to apply the two methods seen in chapter 3 and 4.

From the first method seen in chapter 3, given a desired trajectory, a manipulator, we are able to calculate the torque necessary to follow the desired trajectory and therefore with a single manipulator to be able for example to move different objects without resorting to the manipulator model but thanks to a previously calculated Regressor $\bar{Y}_o$ of the manipulator that can be done by an identification procedure (Open-Loop Identification for our case).

The second method seen in Chapter 4 allows us to calculate the torque necessary to follow the desired trajectory as sums of two contributions, one τ_m which is that necessary to follow the desired trajectory without an object at the end-effector and the other $\bar{\tau}_o$ like the torque to add to the first to compensate/balance the effects due to the object.

In the following chapters, we will see if these methods can be applied to an elastic joint manipulator or soft manipulator and if possible, see if the results are good enough as for a rigid manipulator.

Chapter 6

Elastic Joint Manipulators

6.1 Model Description

The dynamic model of an elastic joint manipulator [6][2] can be described by the following equations system

$$M(q(t))\ddot{q}(t) + C(q(t), \dot{q}(t))\dot{q}(t) + G(q(t)) + K(t)(q(t) - \theta(t)) = 0, \quad (6.1)$$

$$J\ddot{\theta}(t) + D\dot{\theta}(t) - K(t)(q(t) - \theta(t)) = \tau(t), \quad (6.2)$$

where equation (6.1) describes the link dynamics, $M(q(t))$, $C(q(t), \dot{q}(t))$ and $G(q(t))$ are described in section (2.1). $q(t)$ are the joint angles, with their time derivatives $\dot{q}(t)$, $\ddot{q}(t)$, $\dot{\dot{q}}(t)$, $\ddot{\dot{q}}(t)$. Equation (6.2) describes the motor dynamics with the Inertia, Viscous friction, and Stiffness matrices of the motors are respectively J , D and $K(t)$, $\tau(t)$ the actuation torque. $\theta(t)$ are the motor angles, with their time derivatives $\dot{\theta}(t)$, $\ddot{\theta}(t)$.

From the equations above, we try to capture a reality that the rigid model do not consider, the fact that the motor angle is not exactly the joint angle since there is transmission delay due to the manipulator components such as transmission belts, pulleys, long shafts and so for a better control of manipulators we need to include those parameters which can be expressed in terms of J , D , $K(t)$ and also the motor angles.

We can see that setting $q(t) = \theta(t)$, and summing (6.1) and (6.2) we obtain

$$(M(q(t)) + J)\ddot{q}(t) + (C(q(t), \dot{q}(t)) + D)\dot{q}(t) + G(q(t)) = \tau(t) \quad (6.3)$$

which is similar to the rigid manipulator model. Another observation can be done, setting $J = D = 0$ in (6.2), we obtain

$$q(t) = \theta(t) - K(t)^{-1}\tau(t) \quad (6.4)$$

for simplicity just to understand the idea, we suppose a mono-dimensional equation and we can see that for $K(t)$ which tends to infinity the second term in (6.4) tends to zero and $q(t) = \theta(t)$ which is the assumption done for a rigid body

manipulator. So we can consider all manipulators with a high stiffness as rigid or even when $K(t)$ is much larger than J and D . For $K(t)$ which tends to zero, means that the joint angles $q(t)$ tends to $-K(t)^{-1}\tau(t)$ and is independent of the actuators which is not physically possible. This was just to show that Elastic Joint model described in (6.1) and (6.2) is a more general case of manipulator. We will assume that:

H1) J, D and $K(t)$ are diagonal matrix,

H2) $K(t) = K$ is constant.

With the second hypothesis (H2), we talk about Series Elastic Actuators (SEA) [4] to which we dedicate the next chapter.

Chapter 7

Manipulator with Series Elastic Actuators (SEA)

7.1 Model Description

For the sake of clarity, we will drop the dependency on time t . Assuming to deal with an SEA Manipulator, the dynamic model of a serial manipulator with N joints can be

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + K(q - \theta) = 0 \quad (7.1)$$

$$J\ddot{\theta} + D\dot{\theta} - K(q - \theta) = \tau \quad (7.2)$$

It's possible to write the manipulator dynamics as in (2.3)? Here we propose and intuitive/simple way to do it.

Starting with the motor equation (7.2) which is a second order linear time invariant system, thanks to hypothesis (H1) can be rewritten as

$$Y_{\ddot{\theta}}\pi_J + Y_{\dot{\theta}}\pi_D - Y_{q\theta}\pi_K = \tau \quad (7.3)$$

where $Y_{\ddot{\theta}}$, $Y_{\dot{\theta}}$ and $Y_{q\theta}$ are $n \times n$ matrices, and the inertia, viscous damping and stiffness parameters vectors π_J , π_D and π_K are column vectors of dimension n .

$$Y_{\ddot{\theta}} = \text{diag}(\ddot{\theta}), \quad Y_{\dot{\theta}} = \text{diag}(\dot{\theta}), \quad Y_{q\theta} = \text{diag}(q - \theta),$$

$$\pi_J = [j_1 \ j_2 \ \dots \ j_n]^T, \ \pi_D = [d_1 \ d_2 \ \dots \ d_n]^T, \ \pi_K = [k_1 \ k_2 \ \dots \ k_n]^T.$$

From equations (2.1), (2.2), (2.3) and (7.3), the link equation can be rewritten as :

$$Y_m(q, \dot{q}, \ddot{q})\pi_m + Y_o(q, \dot{q}, \ddot{q})\pi_o + Y_{q\theta}\pi_K = 0 \quad (7.4)$$

So the dynamic model can be rewrite as:

- Link equation : (7.4)

- Motor equation : (7.3)

Summing up (7.4) and (7.3) we obtain

$$Y_m(q, \dot{q}, \ddot{q})\pi_m + Y_{\ddot{\theta}}\pi_J + Y_{\dot{\theta}}\pi_D + Y_o(q, \dot{q}, \ddot{q})\pi_o = \tau \quad (7.5)$$

putting the terms in the following way

$$\begin{bmatrix} Y_m(q, \dot{q}, \ddot{q}) & Y_{\ddot{\theta}} & Y_{\dot{\theta}} \end{bmatrix} \begin{bmatrix} \pi_m \\ \pi_J \\ \pi_D \end{bmatrix} + Y_o(q, \dot{q}, \ddot{q})\pi_o = \tau \quad (7.6)$$

Pay attention that in (7.5) and (7.6), the stiffness parameters vector π_K has been canceled. Finally we arrive to the desired form

$$Y_M(q, \dot{q}, \ddot{q}, \dot{\theta}, \ddot{\theta})\pi_M + Y_o(q, \dot{q}, \ddot{q})\pi_o = \tau_M + \tau_o = \tau \quad (7.7)$$

which is similar to the form (2.3).

Y_M and π_M are the Regressor and the Dynamic parameters vector of a manipulator with Series Elastic Actuators.

We can finally conclude that both methods 1 and 2 can be applied to manipulators with Series Elastic Actuators which is a particular class of Elastic Manipulator as mentioned above.

We have shown that the torque of a SEA manipulator can be written as the sum of two contributions as in the rigid case. Now let's see how do we compute the required or desired torque τ_d given the desired trajectory q_d ?.

7.2 Torque Computation

A better description can be found in the papers[6] [3]. For the sake of simplicity, we will use the matrix form and set : $C(q, \dot{q})\dot{q} + G(q) = N(q, \dot{q})$.

We find the expression of theta θ from link equation (7.1), differentiate it with respect to time an obtain the expression of $\dot{\theta}$, once again differentiate with respect to time and obtain the expression of $\ddot{\theta}$.

So we get

$$\theta = q + K^{-1}(M(q)\ddot{q} + N(q, \dot{q})) \quad (7.8)$$

$$\dot{\theta} = \dot{q} + K^{-1}(M(q)\dot{\ddot{q}} + \dot{M}(q)\ddot{q} + \dot{N}(q, \dot{q})) \quad (7.9)$$

$$\ddot{\theta} = \ddot{q} + K^{-1}(M(q)\ddot{\ddot{q}} + 2\dot{M}(q)\dot{\ddot{q}} + \ddot{M}(q)\ddot{q} + \ddot{N}(q, \dot{q})) \quad (7.10)$$

Now let's insert (7.8), (7.9) and (7.10) in the motor equation (7.2) to obtain

$$\begin{aligned} \tau = & JK^{-1}(M(q)\ddot{q} + 2\dot{M}(q)\dot{q} + (K + \ddot{M}(q))\ddot{q} + \ddot{N}(q, \dot{q})) + \\ & DK^{-1}(M(q)\dot{q} + \dot{M}(q)\ddot{q} + K\dot{q} + \dot{N}(q, \dot{q})) - (M(q)\ddot{q} + N(q, \dot{q})) \end{aligned} \quad (7.11)$$

Finally we obtain the whole system dynamics express in terms of the link variables q and their derivatives. Therefore given the desired trajectory q_d , the necessary feed-forward torque is compute using the equation (7.11) evaluated in q_d .

$$\tau_d = \tau(q_d, \dot{q}_d, \ddot{q}_d, \dot{\ddot{q}}_d, \ddot{\ddot{q}}_d) \quad (7.12)$$

7.3 Evaluation and Results

For the evaluation, We will use the same Criteria as in section 3.2.2 using (3.7) and (3.8). For the results, the same data are used for the simulation as in chapter 3 and 4. The last parameters to set are:

$j_i = 0.001$, $d_i = 0.01$, $k_i = 50$, $i = 1 \dots n$ where n is the number of Joints.

As before, object 5 is used for the falsification.

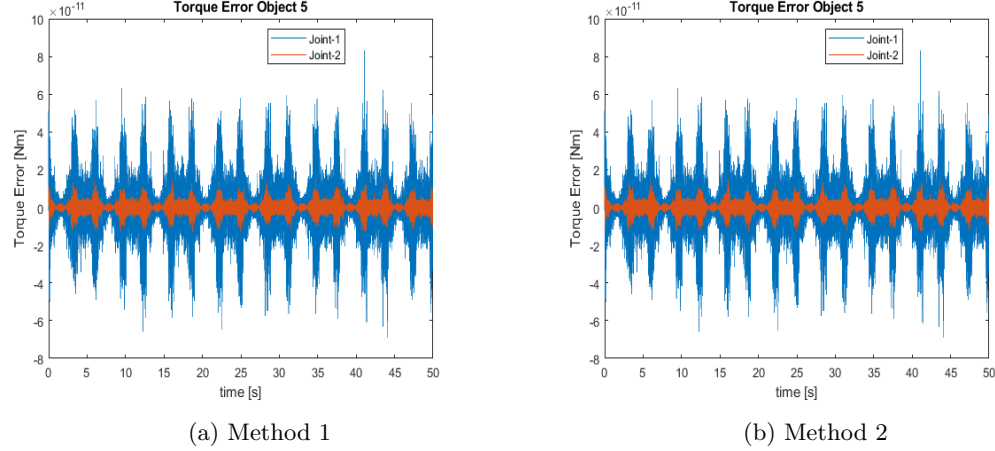
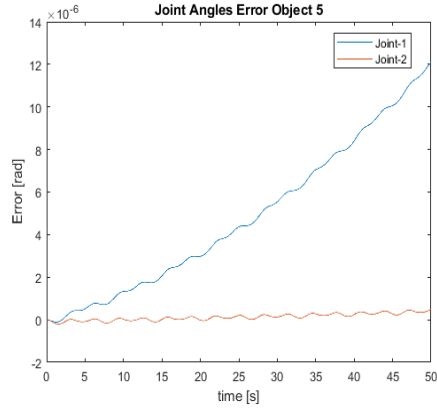
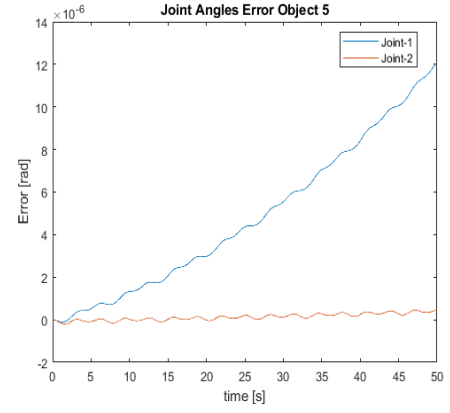


Figure 7.1: Torque Error

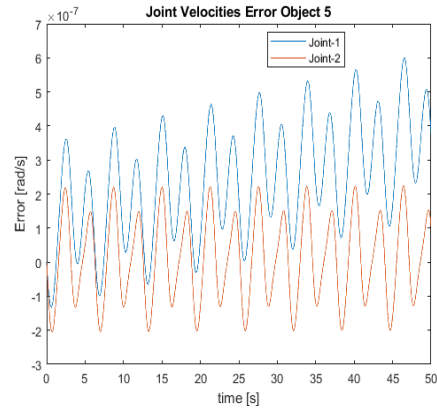


(a) Method 1

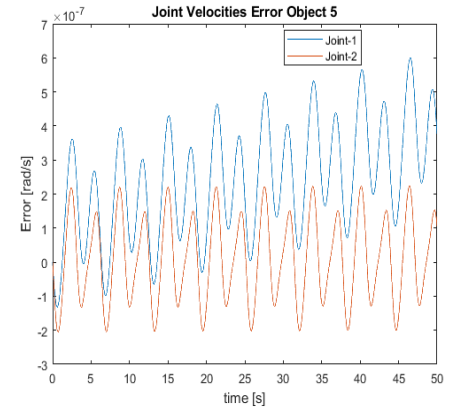


(b) Method 2

Figure 7.2: Joint Angles Error



(a) Method 1



(b) Method 2

Figure 7.3: Joint Velocities Error

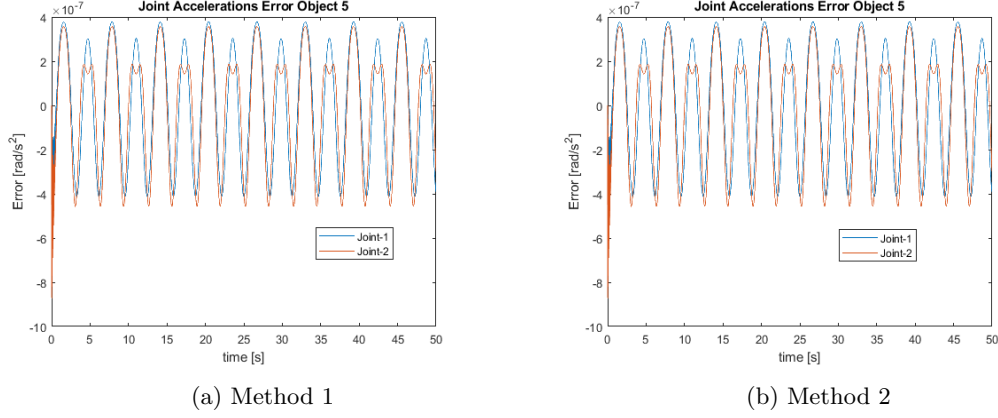


Figure 7.4: Joint Accelerations Error

7.4 Conclusion

We can easily conclude that even for an SEA manipulator the proposed methods are applicable and give good results, but so similar that they do not allow us from the graphs to determine which is better than the other.

As for the two types of manipulators studied, in joint accelerations and torque, there is a degradation of performance since the errors are greater for the sea case but despite this the desired trajectory is pursued with an error equal to the rigid case.

7.5 Remarks

From the equations (2.3), (2.7), (4.5), (4.6), (7.7), the fact of being able to write the torque as the sum of the torque needed to move the empty manipulator (τ_m) and that due to the object at the end-effector (τ_o), does not only allow us to think about the generalization of objects as we have done so far but perhaps it could also allow us to (method 2): given the same desired trajectory of an experiment previously carried out, given a new manipulator to calculate only the torque as

$$\tau = {}^i \tau_m + \tau_o \quad (7.13)$$

Obviously there will be constraints to be respected:

- The number of joints (N) must be equal.
- The type of joints (revolute, prismatic) must be corresponding and in the same order from the base to the end effector.
- The length (a_i) of the joints must be equal.

Or more simply that they must have the same kinematics.

For our case, the last constraint could be overcome by considering (a_i) as a dynamic parameter and so the regressor will be a function only of the joint angles q and its derivatives. And so for now, for feasible trajectories, we can say that it can be done between rigid-rigid and rigid-sea manipulators.

After these assumptions, for verification we can compute as in equation (4.5) the difference between object torque computed from the rigid manipulator (${}^r\tau_o$) and the object torque computed from the sea manipulator (${}^s\tau_o$) and expect it to be null.

We performed the test with object 5, and the result is shown in figure 7.5. As we can see, it is not what we expected.

To try to explain this, we think that the mathematical steps taken to obtain equation (7.7) which guarantees the application of methods 1 and 2 are correct, but they have the defect of canceling part of the dynamics of the manipulator and more precisely the part which contains the stiffness ($K(q - \theta)$). In the real world, dependence on stiffness persists.

If mathematically we can cancel the stiffness and think that there is no more dependence, means that there are operations in the physical world that can degrade the stiffness of the manipulator or in the worst case cancel it which as seen above in equation (6.4) is to be avoided.

To summarize, in the case of a sea manipulator, the link and motor equation depend intrinsically on stiffness K and if we try to eliminate this dependence, we get into trouble. This can be resumed in $q = q(\theta(K))$ and therefore understand the equation (6.4) accordingly, otherwise we could make erroneous assumptions as happens with the assumption made above in equation (7.13).

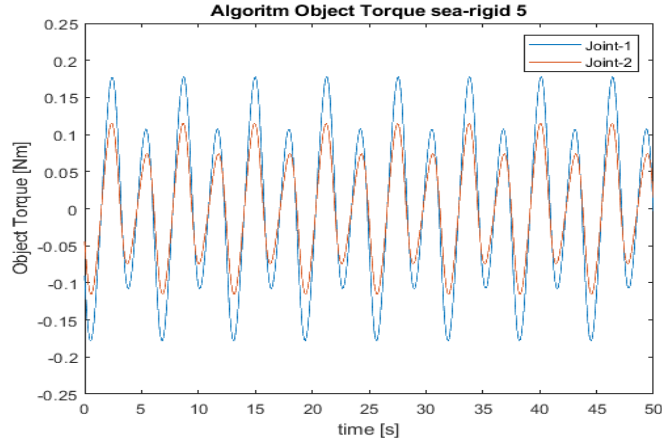


Figure 7.5: Object Torque difference between rigid and sea manipulator.

Chapter 8

SEA Manipulator with Object as a Link

With everything we've seen so far, it's easy to compute the dynamics of a sea manipulator with Object as a Link (see chapters 5 and 7).

8.1 Results

When we apply both methods, we get the results in figures 8.1, 8.2, 8.3, 8.4.

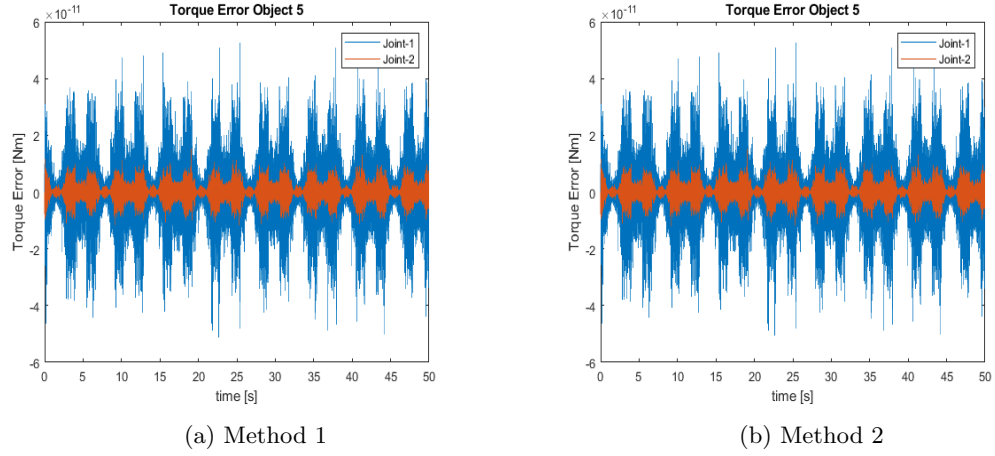
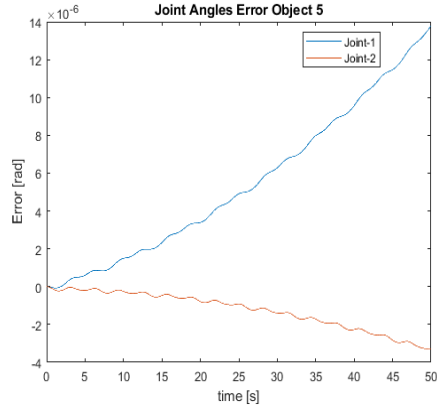
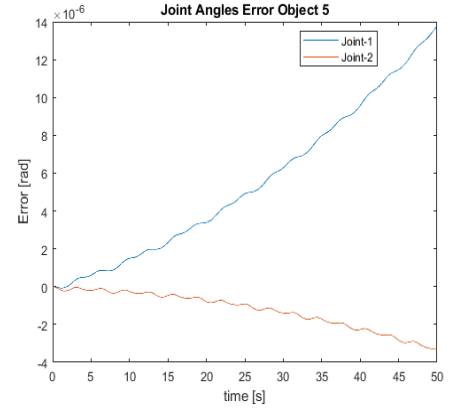


Figure 8.1: Torque Error

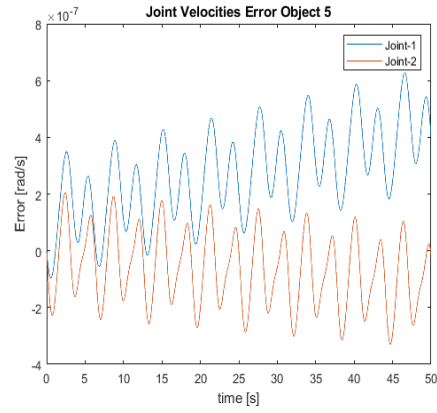


(a) Method 1

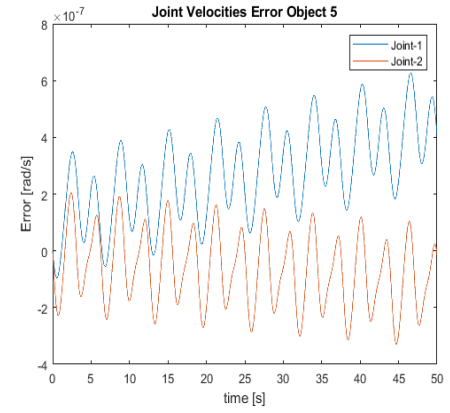


(b) Method 2

Figure 8.2: Joint Angles Error

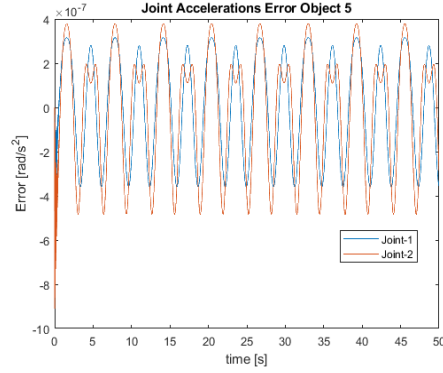


(a) Method 1

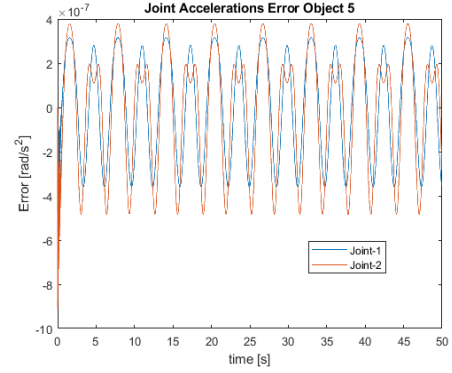


(b) Method 2

Figure 8.3: Joint Velocities Error



(a) Method 1



(b) Method 2

Figure 8.4: Joint Accelerations Error

8.2 Conclusion

So far we have seen that the two methods proposed for our study give more or less the same results from the graphs and therefore we are not able by all what we have done to say which is better than the other. For this reason we will test these methods in the next chapter

Chapter 9

Methods Comparison

In this chapter we want to investigate the methods to know or decide which is better than the other and in which situation. So we did the two experiments that follow. For both simulations, we used a rigid manipulator with object described as a link (see chap 5), we used object 0 to 3 to build the numerical regressor and object 4 and 5 (see table 5.2) to do the test.

9.1 Experience 1 : Uncertainty on the Object Dynamic Parameters

Having available the numerical regressor and the parameters of the object, we can compute the torque necessary to track the desired trajectory $q_d(t)$. We therefore hypothesized to have uncertainty on the parameters of the object of $\pm 2\%$. We compute the torque to be applied to the joints with the uncertain parameters and the regressor according to the equation 3.5 for the method 1 and according to the equations 4.5 and 4.6 for the method 2. We then investigated the desired torque error $\tau_d - \tau_{A_i}$, the tracking error $q_d - q_{A_i}$ (see equation 3.8 a) and the difference of the values given by each method $\tau_{A_1} - \tau_{A_2}$ and $q_{A_1} - q_{A_2}$.

9.2 Experience 2 : Singular Configuration of the Object Dynamic Parameters Matrix

From equation 3.4 and equation 4.4 we know that in order for the methods to be applied, the matrix of the parameters of the objects used in the identification must be invertible or it must be far from the configuration of singularity. We will therefore exploit this property to test the methods, imposing that this matrix is in a condition of singularity. For this, it is sufficient to impose that all the objects used for the construction of the regressor have equal center of mass (l_o) distance from the end-effector and so the masses row will be linearly dependent with the row of the first moments of inertia ($m_o l_o$). Then we will try to remove

this matrix from the singularity configuration by applying a diagonal loading in order to have a torque to apply to the joints and evaluate the two methods as in the first experiment.

9.3 Results

No graph is reported as for each experiment we have done, the data obtained are exactly the same and when we make the difference between the data of each method, we find the horizontal line with ordinate in zero this means that method 1 and 2 are indistinguishable and therefore we can use one or the other with no preferences. However, it's what we can deduce from the experiments made and also from the results obtained in chapters 3, 4, 5, 7 and 8.

One thing that could help us to choose one over the other may be the computational cost, and method 1 seems more expensive than method 2 (number of operations).

9.4 Conclusion

We saw two different ways of representing a manipulator, the first one in chapter 2 which is good when the centre of mass of the object coincide with the end-effector position but it's not always true. The second way in chapter 5 is better than the first since it integrates more object parameters which can result convenient for us.

For the methods used to computed the feed-forward torque, the initial proposed method was method 1 and method 2 was just a second way to achieved the same goal and this second method help us for a better understanding of manipulators behaviour.

Therefore, in the rest of the work, we will use the dynamic model with object as a link (see chapter 5) and for the methods, we will use the method 1 since we are not in need of fast computation or any other particular requirements.

Part III

Application

Chapter 10

Feed-forward Control Law

As described in chapter 3, we want to apply the feed-forward control law in equation (3.5)

$$\tau = \tau_A = \bar{Y}_o \bar{\pi} \quad (10.1)$$

10.1 Manipulator Description

For the sake of simplicity we decide to use a 3-DOFs manipulator which Denavit-Hartenberg parameters are in table 10.1 and the manipulator can be represent as in figure 10.1.

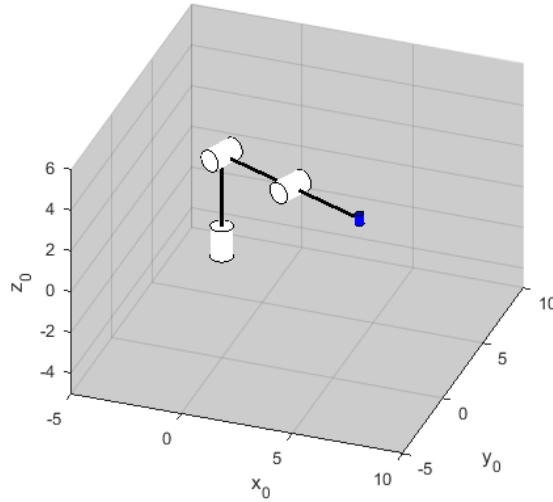


Figure 10.1: Three-Link Arm Manipulator

i	a_i	α_i	d_i	θ_i
1	0	$\pi/2$	a_1	q_1
2	a_2	0	0	q_2
3	a_3	$\pi/2$	0	q_3

Table 10.1: Denavit-Hartenberg Parameters

Parameters	Link 1	Link 2	Link 3
$l(m)$	0.5	0.5	0.5
$a(m)$	1	1	1
m_l (kg)	50	50	50
m_m (kg)	5	5	5
$I_{xx}(kg.m^2)$	10	0	0
$I_{yy}(kg.m^2)$	0	10	10
$I_{zz}(kg.m^2)$	0	10	10
$I_m(kg.m^2)$	0.01	0.01	0.01
$m_e(kg)$	0	0	5
$I_e(kg.m^2)$	0	0	0.01
k_r	100	100	100

Table 10.2: Manipulator Parameters Values

The data in table 10.2 are the manipulator parameters values:

10.2 Object Dynamic Parameters

First of all, we will like to give a more detailed description of (5.4), the object dynamic parameters vector

$$\pi_o = [m_o \quad m_o l_o \quad I_o + m_o l_o^2]^T$$

which can be written as

$$\pi_o = [m_o \quad m_o l_{o_x} \quad m_o l_{o_y} \quad m_o l_{o_z} \quad I_{o_x} + m_o l_{o_x}^2 \quad I_{o_y} + m_o l_{o_y}^2 \quad I_{o_z} + m_o l_{o_z}^2]^T \quad (10.2)$$

As we can see is a vector of dimension 7, and therefore, ${}^i\bar{\pi}_o = \begin{bmatrix} 1 \\ {}^i\pi_o \end{bmatrix}$ is a vector of dimension 8.

So for the method to be apply, we need 8 objects.

In this work, instead of 7 we will reduce the dimension of the object parameters vector to 5 by setting

$$\pi_o = [m_o \quad m_o l_{o_x} \quad I_{o_x} + m_o l_{o_x}^2 \quad I_{o_y} + m_o l_{o_y}^2 \quad I_{o_z} + m_o l_{o_z}^2]^T \quad (10.3)$$

<i>Object(i)</i>	<i>m</i> (kg)	<i>a_o</i> (m)	<i>w</i> (m)	<i>h</i> (m)
0	0.0	0.0	0.00	0.00
1	10	0.1	0.05	0.03
2	20	0.2	0.20	0.06
3	30	0.3	0.15	0.10
4	40	0.4	0.13	0.20
5	50	0.5	0.25	0.16
6	60	0.6	0.30	0.2
7	70	0.7	0.46	0.35
8	80	0.8	0.40	0.32
9	90	0.9	0.45	0.30
10	100	1.0	0.25	0.33

Table 10.3: Object Parameters Values

As we can see, we use only one component of the first moment of inertia and therefore we will need only 6 objects for the torque identification. The set of objects used for our study are in table 10.3

Where m, a_o, w, h are respectively the mass, length, width and height of the object. COM is $L_{o_x} = a_o/2, L_{o_y} = w/2, L_{o_z} = h/2$

For the inertia moment, we compute it as

$$I_o = \frac{m}{12} \begin{bmatrix} w^2 + h^2 & 0 & 0 \\ 0 & a_o^2 + h^2 & 0 \\ 0 & 0 & a_o^2 + w^2 \end{bmatrix}$$

At this point, we define object 0 to object 5 those we will use to construct the numerical regressor and object 7 to object 10 for the test.

10.3 Desired Trajectory

We define the general desired trajectory as

$$q_i(t) = \frac{A}{i} \left\{ c_1 \left(\frac{t}{T} \right)^1 + c_2 \left(\frac{t}{T} \right)^2 + c_3 \left(\frac{t}{T} \right)^3 + c_4 \left(\frac{t}{T} \right)^4 + c_5 \left(\frac{t}{T} \right)^5 + c_6 \left(\frac{t}{T} \right)^6 \right\}, i = 1 \dots n$$

Where A, c_j are coefficients that we can easily use to generate different trajectories, T the simulation time s and n the number of joints of the manipulator.

10.4 Rigid Manipulator Results

10.4.1 First trajectory

The first trajectory is obtain by setting :

$A = \pi$, $c_1 = 0$, $c_2 = 0$, $c_3 = 5$, $c_4 = -10$, $c_5 = 5$, $c_6 = 0$ and $T = 10s$. The desired position is shown in figure 10.2.

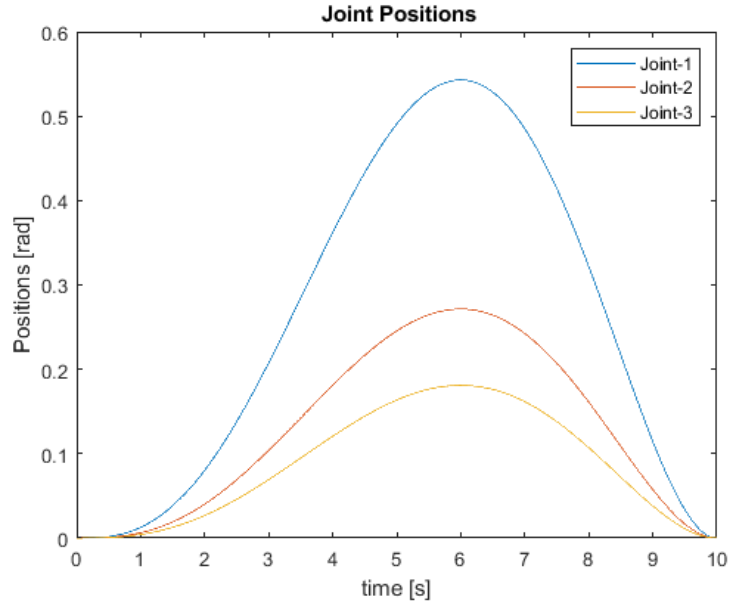


Figure 10.2: Rigid Manipulator First Desired Trajectory

For this trajectory, the results when apply to object 7 and object 10 are shown in figures 10.3 and 10.4.

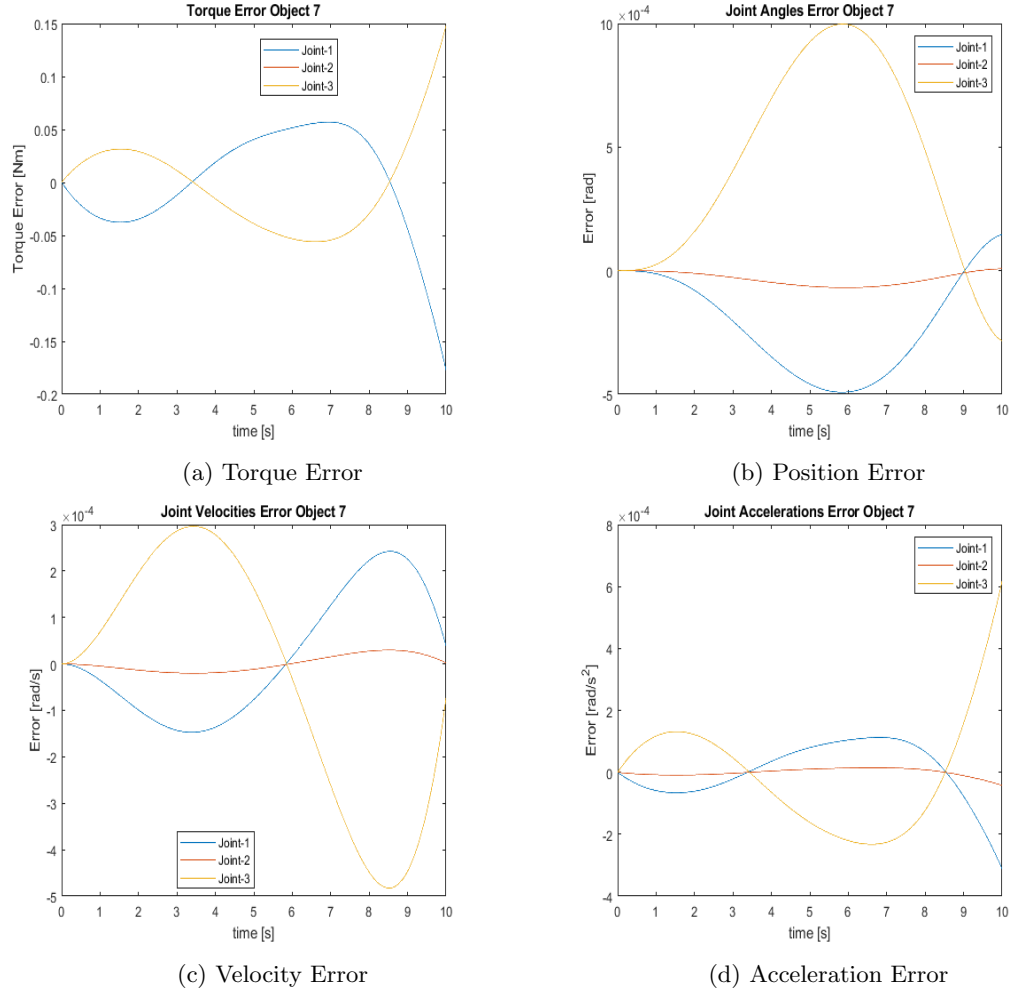
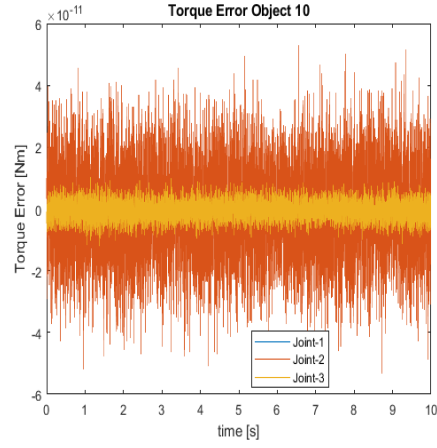
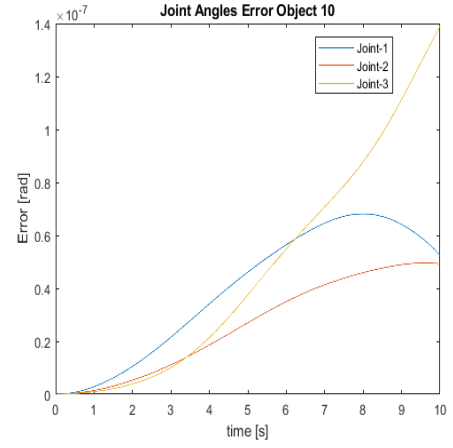


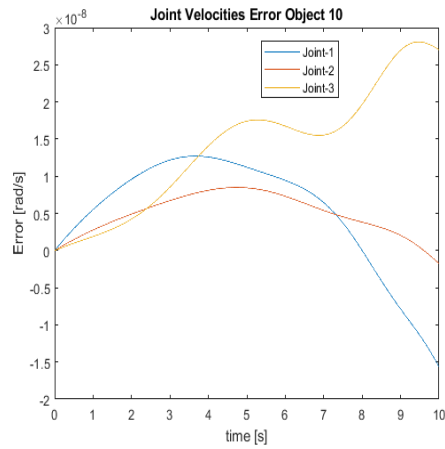
Figure 10.3: Tracking Errors on first desired trajectory with a rigid manipulator and Object 7



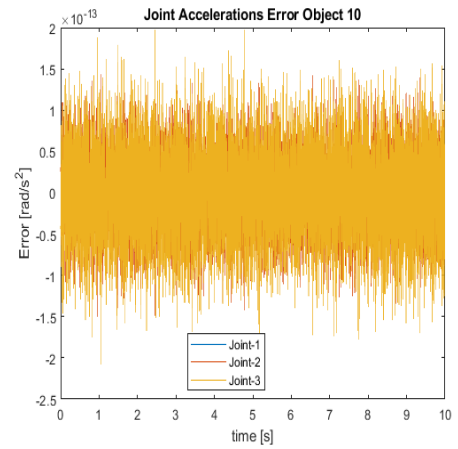
(a) Torque Error



(b) Position Error



(c) Velocity Error



(d) Acceleration Error

Figure 10.4: Tracking Errors on first desired trajectory with a rigid manipulator and Object 10

10.4.2 Second trajectory

The second trajectory is obtain by setting :

$A = \pi$, $c_1 = 0$, $c_2 = 0$, $c_3 = 10$, $c_4 = -15$, $c_5 = 6$, $c_6 = 0$ and $T = 10s$. The desired position is shown in figure 10.5.

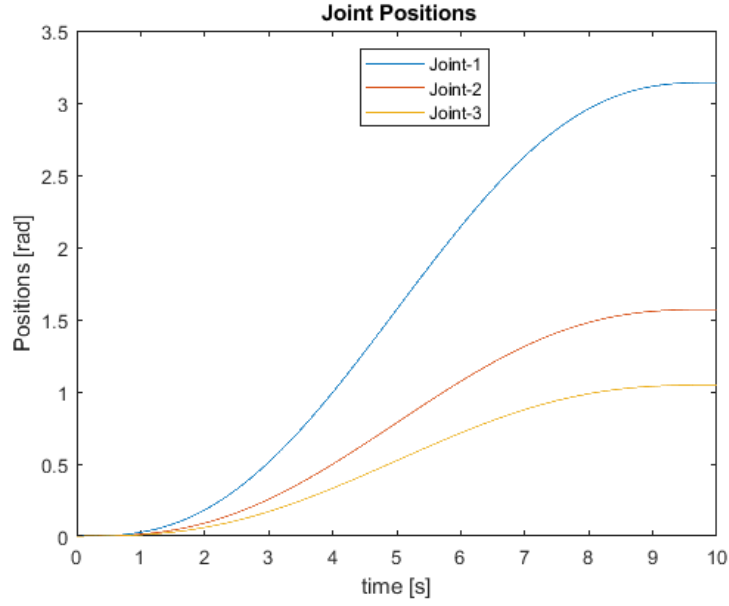
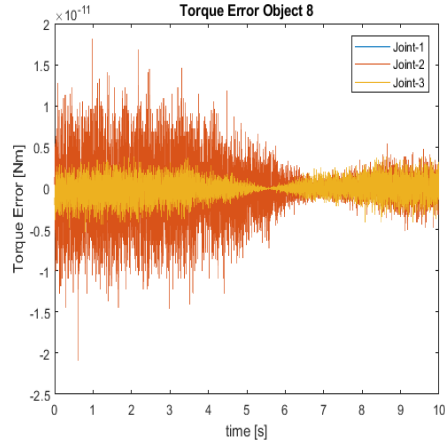
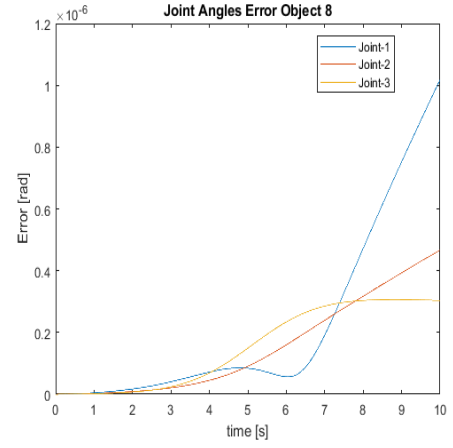


Figure 10.5: Rigid Manipulator Second Desired Trajectory

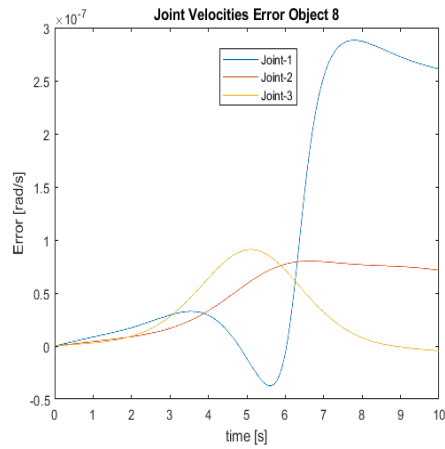
For this trajectory, the results when apply to object 8 and object 9 are shown in figure 10.6 and figure 10.7.



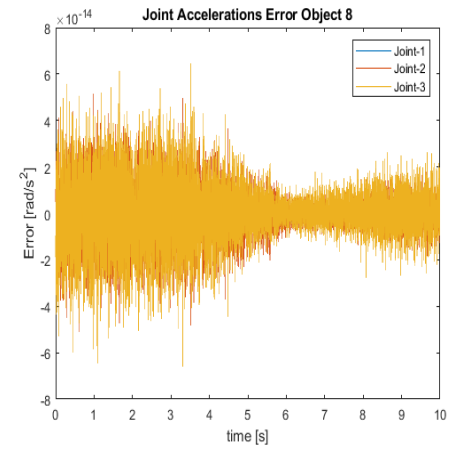
(a) Torque Error



(b) Position Error

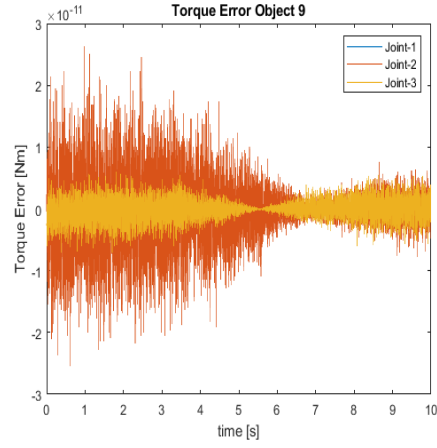


(c) Velocity Error

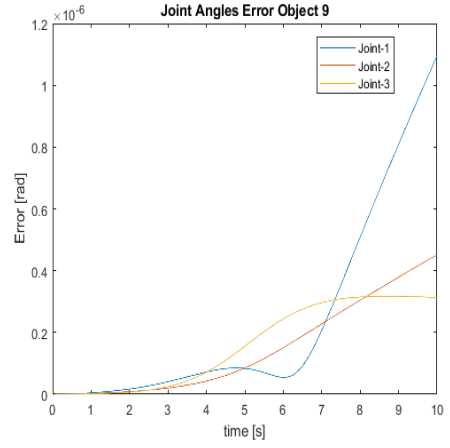


(d) Acceleration Error

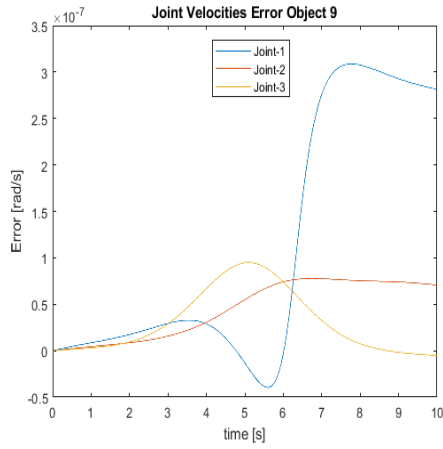
Figure 10.6: Tracking Errors on second desired trajectory with a rigid manipulator and Object 8



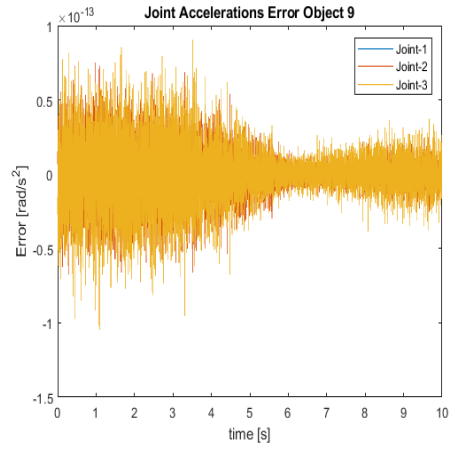
(a) Torque Error



(b) Position Error



(c) Velocity Error



(d) Acceleration Error

Figure 10.7: Tracking Errors on second desired trajectory with a rigid manipulator and Object 9

10.5 SEA Manipulator

The Motor parameters are the following:

$j_i = 0.001$, $d_i = 0.01$, $k_i = 50$, $i = 1 \dots n$ where n is the number of Joints.

10.5.1 First trajectory

The desired trajectory is the same as in figure 10.5. The results are shown in figure 10.8 and figure 10.9 when apply to object 8 and object 9.

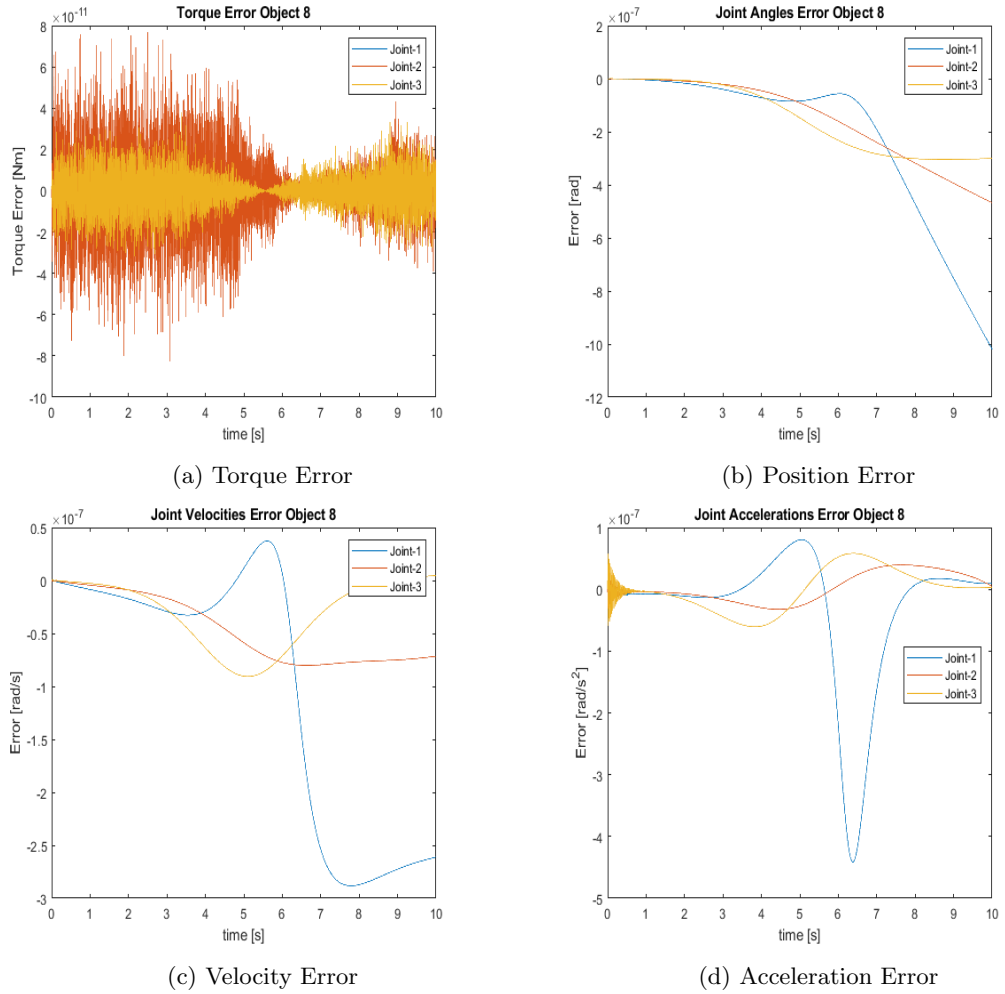


Figure 10.8: Tracking Errors on first desired trajectory with a SEA manipulator and Object 8

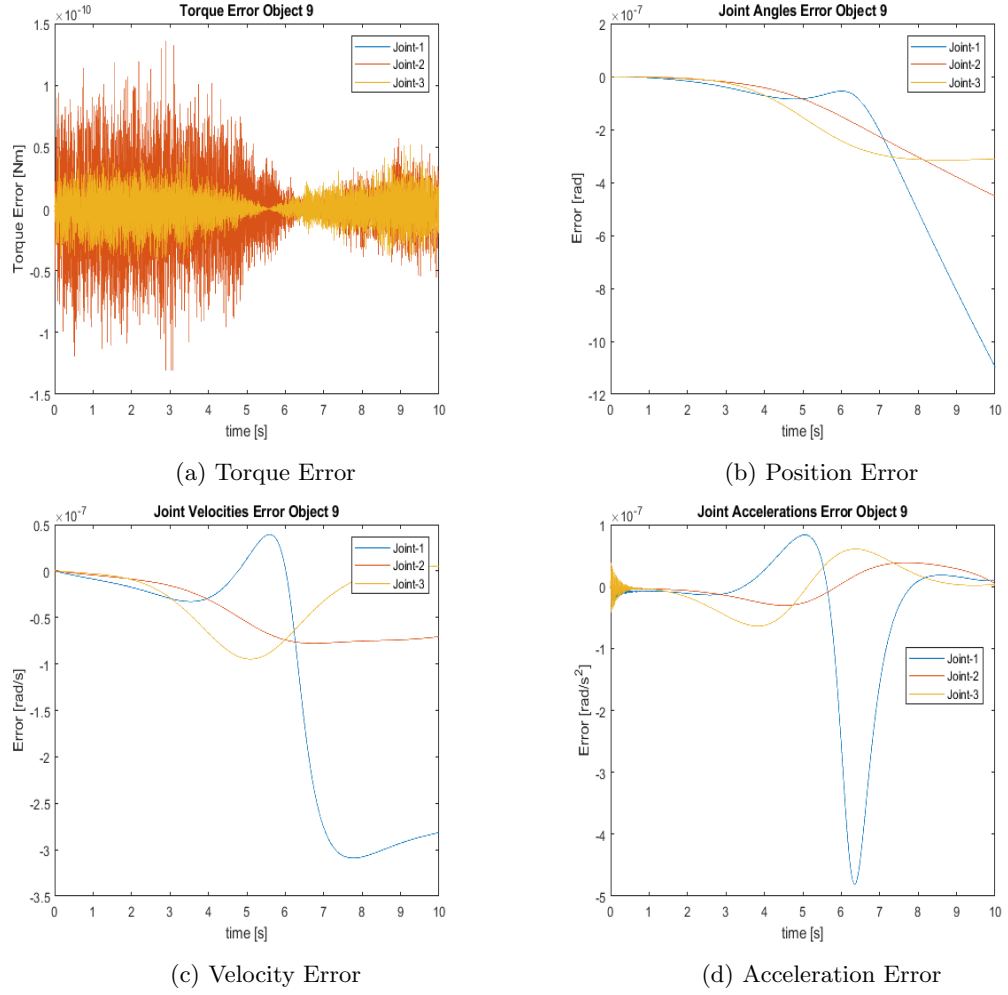


Figure 10.9: Tracking Errors on first desired trajectory with a SEA manipulator and Object 9

10.5.2 Second trajectory

The second trajectory is obtain by setting :

$A = 1$, $c_1 = 0$, $c_2 = 56.6893$, $c_3 = -305.0427$, $c_4 = 574.9919$, $c_5 = -461.6132$, $c_6 = 134.9746$ and $T = 10s$. The desired trajectory is shown in figure 10.10.

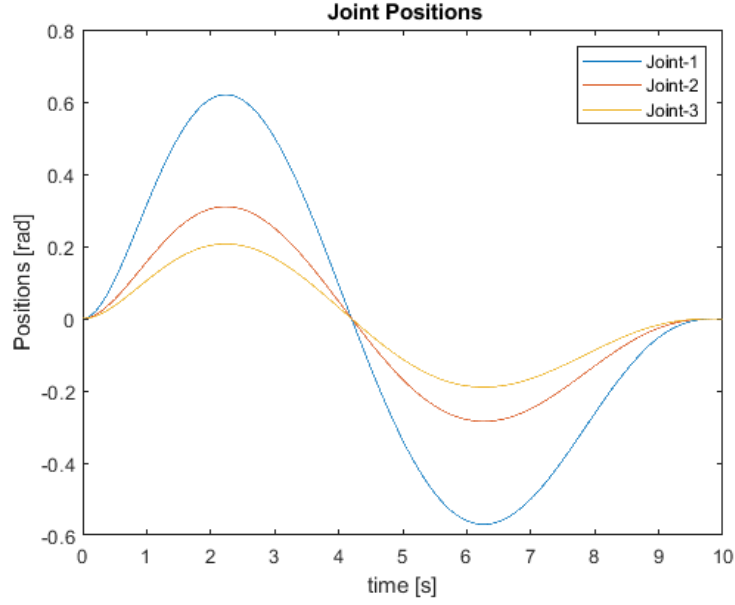


Figure 10.10: SEA Manipulator Second Desired Trajectory

For this trajectory, the results when apply to object 7 and object 10 are shown in figure 10.11 and figure 10.12.

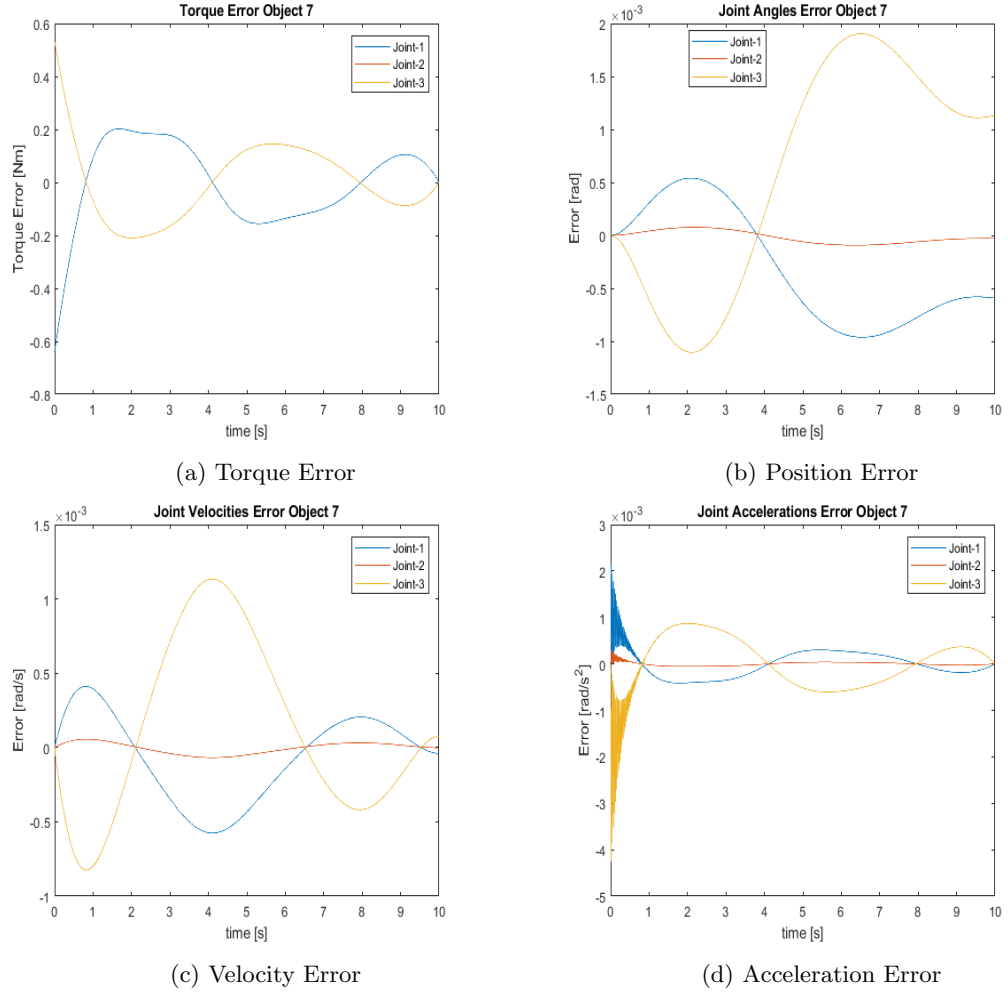
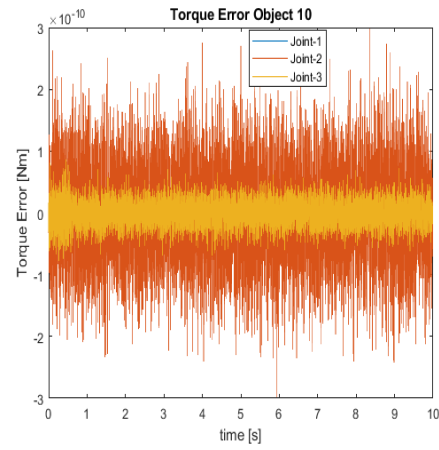
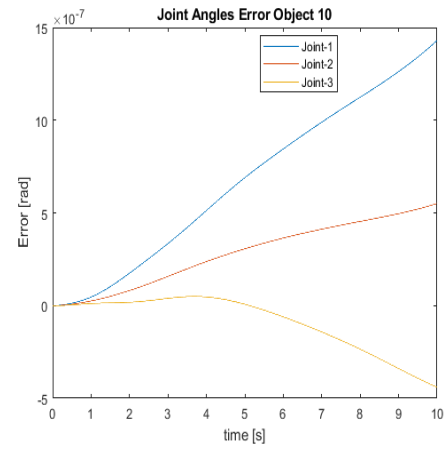


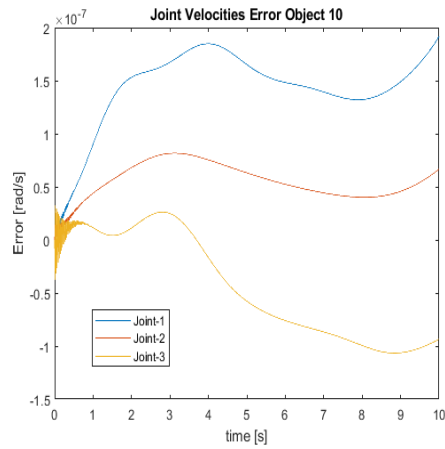
Figure 10.11: Tracking Errors on second desired trajectory with a SEA manipulator and Object 7



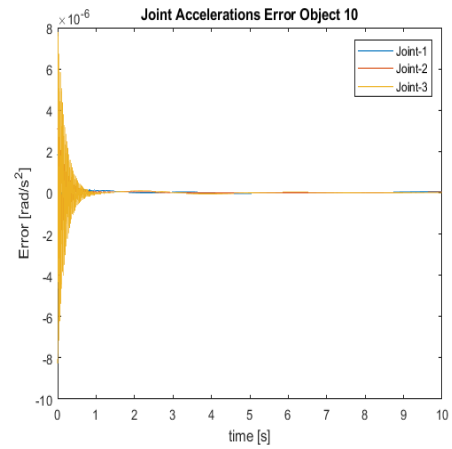
(a) Torque Error



(b) Position Error



(c) Velocity Error



(d) Acceleration Error

Figure 10.12: Tracking Errors on second desired trajectory with a SEA manipulator and Object 10

Chapter 11

Feed-forward/Feedback Control Law

Actually, the feed-forward torque we have used so far is not robust enough in presence of inaccurate initial conditions or uncertainty on object parameters. For this reason we can try to update the previous feed-forward control law by adding a feedback control law which will be a PD control law.

11.1 Rigid Manipulator Results

For a rigid manipulator, assuming link position and velocity available, we can apply the control law in equation (11.1)

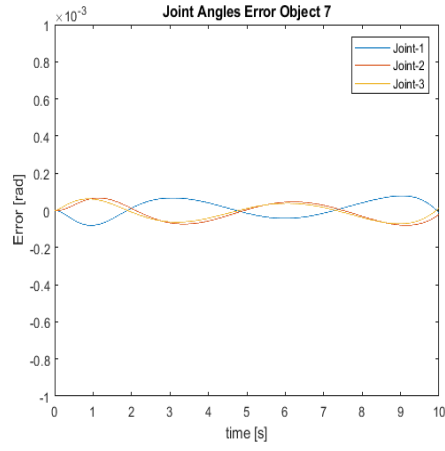
$$\tau = \tau_A + K_p e_q + K_v \dot{e}_q, \quad (11.1)$$

where $K_p > 0$ and $K_v > 0$ for global asymptotic stability.

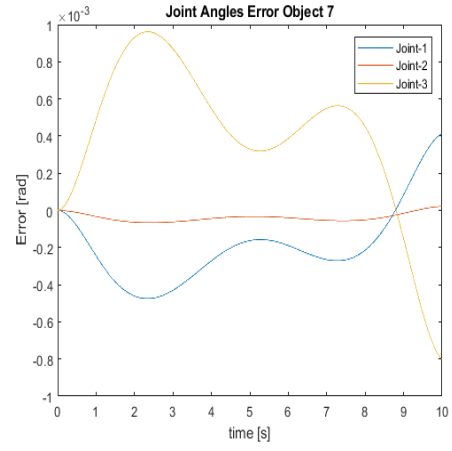
The desired trajectory is obtain by setting :

$A = 1$, $c_1 = 0$, $c_2 = 45.66$, $c_3 = -254.77$, $c_4 = 536$, $c_5 = -490.34$, $c_6 = 963.41$ and $T = 10s$.

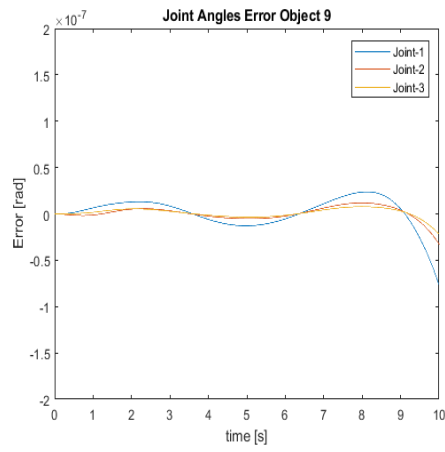
The figure 11.1 shows the position errors when the initial positions and velocities are perfectly known were we can see how the result with feedback is much better. Figure 11.2 shows the tracking performance and figure 11.3 the feed-forward and feedback torque applied when initial positions and velocities are not perfectly known. We used $K_p = 1500$, $K_v = 1000$



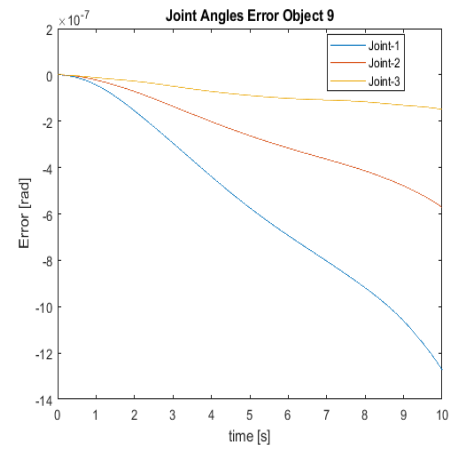
(a) Position Error with feedback



(b) Position Error without feedback

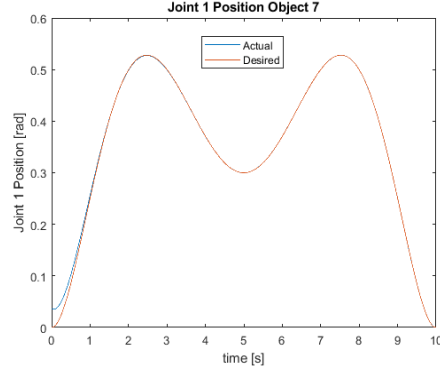


(c) Position Error with feedback

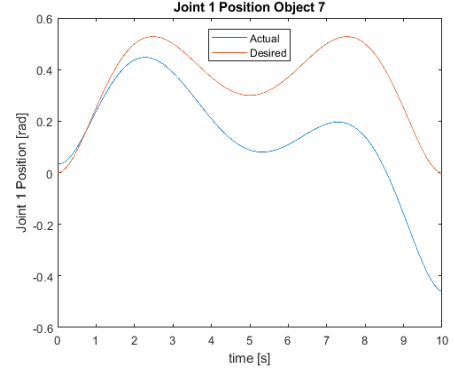


(d) Position Error without feedback

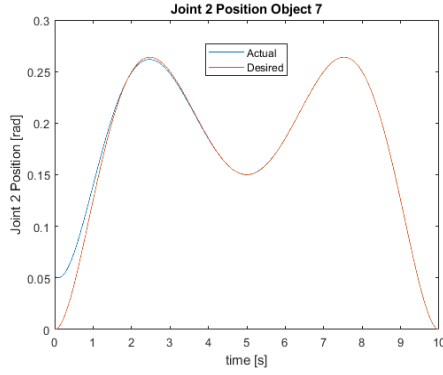
Figure 11.1: Position Errors on Object 7 and 9 With(left) and Without(right) feedback



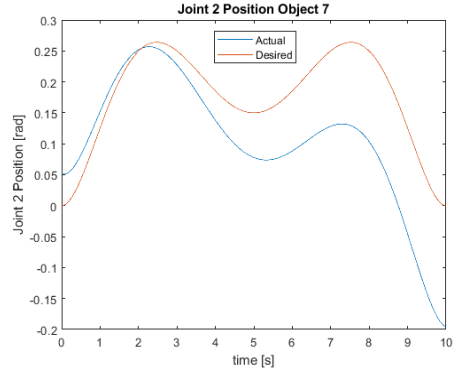
(a) Joint 1



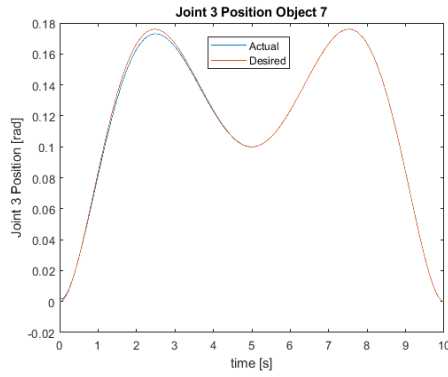
(b) Joint 1



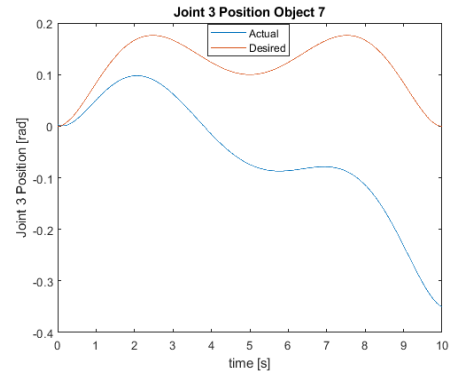
(c) Joint 2



(d) Joint 2



(e) Joint 3



(f) Joint 3

Figure 11.2: Tracking Performance on Object 7 with(left) and without(right) feedback.

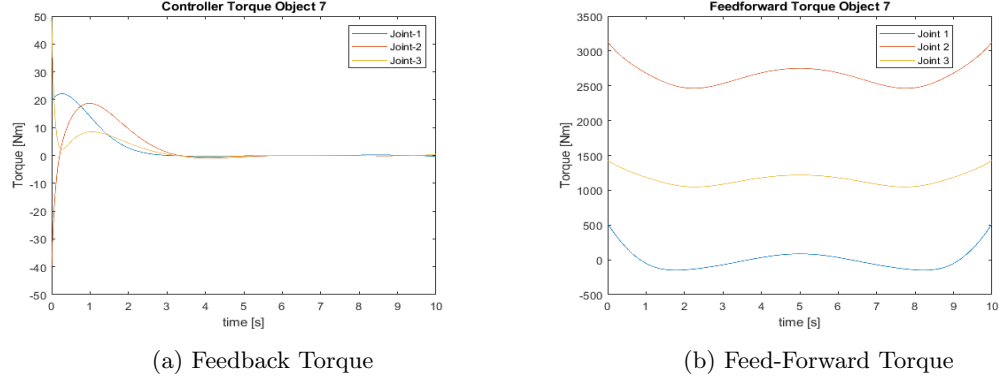


Figure 11.3: Feedback(left) and Feed-forward(right) Torque for object 7.

11.2 SEA Manipulator

The desired trajectory is obtain by setting :

$A = \pi$, $c_1 = 0$, $c_2 = 0$, $c_3 = -14.6026$, $c_4 = 51.3079$, $c_5 = -55.8079$, $c_6 = 19.6026$ and $T = 10s$.

The feed-forward control law when only the link position is inaccurate guarantee what we can call Shape Tracking or Biased Tracking where the bias is the initial position error of the link which remain more or less constant for all the duration of the tracking as shown in figure 11.4 and figure 11.5 (left).

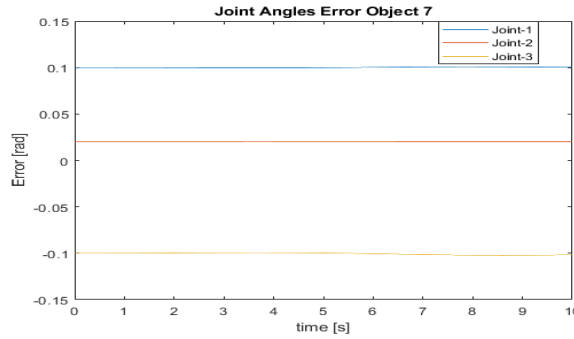
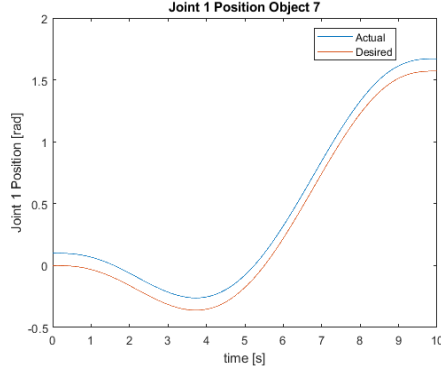
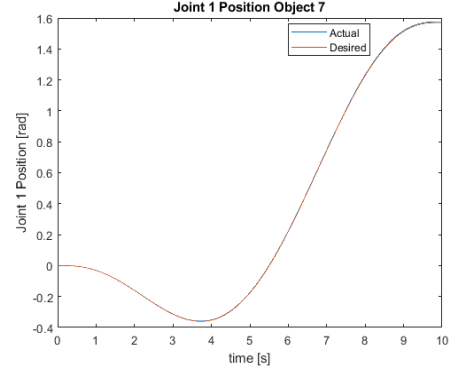


Figure 11.4: Tracking Error for inaccurate link initial position.

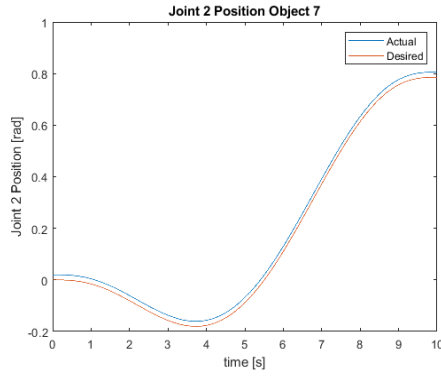
When only the motor initial positions and velocities are inaccurate, as show in figure 11.5(right) the feed-forward continues to guarantee small tracking errors and this for a large range of uncertainty of the motor initial position.



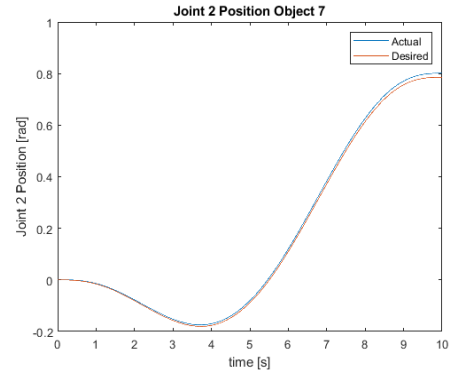
(a) Joint 1



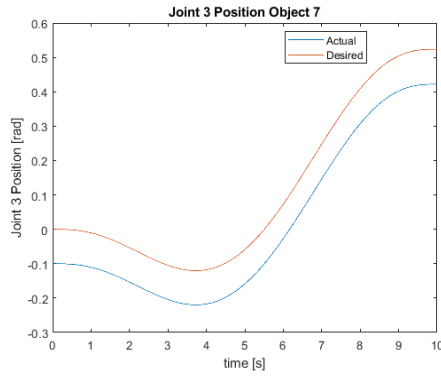
(b) Joint 1



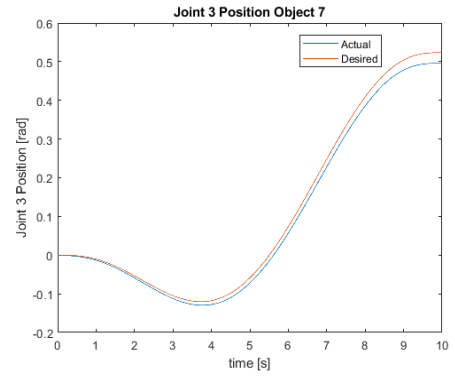
(c) Joint 2



(d) Joint 2



(e) Joint 3



(f) Joint 3

Figure 11.5: Tracking Performance on Object 7 with inaccurate link(left) and motor(right) initial position and velocity.

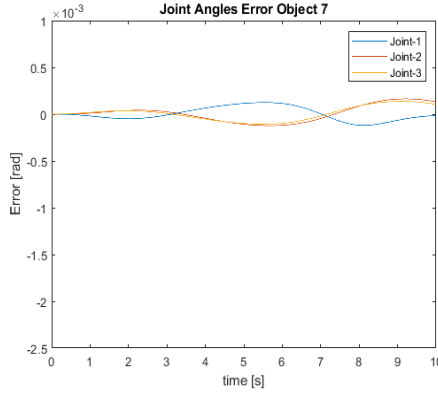
11.2.1 Feedback from Links Positions

So for a SEA manipulator, assuming link position available, we will apply the control law in equation (11.2)

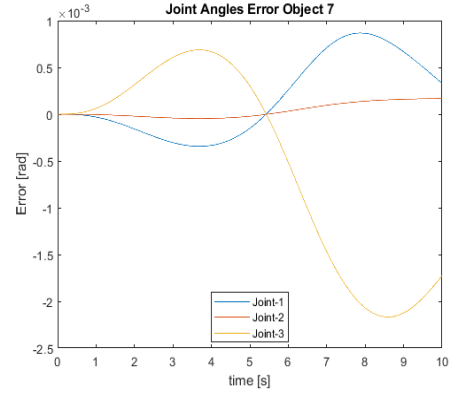
$$\tau = \tau_A + K_p e_q + K_v \dot{e}_q, \quad (11.2)$$

where for $K_p > 0$ and $K_v > 0$ for global asymptotic stability.

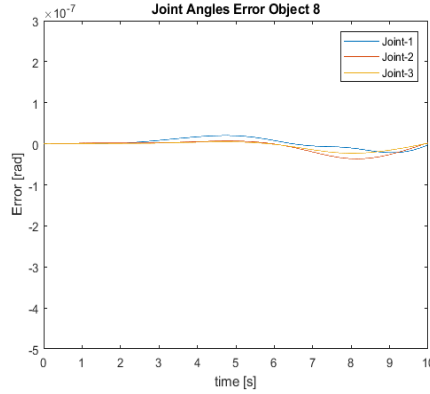
The figure 11.6 shows the position errors when the initial positions and velocities of links and motors are perfectly known were we can see how the result with feedback is much better. Figure 11.7 shows the tracking performance and figure 11.8 the feed-forward and feedback torque applied when initial positions and velocities of the links and motors are not perfectly known. We used $K_p = 1200$, $K_v = 800$.



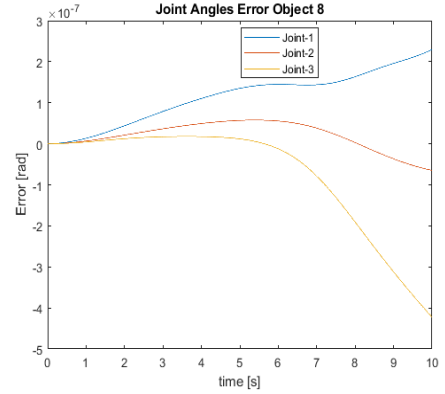
(a) Position Error with feedback



(b) Position Error without feedback

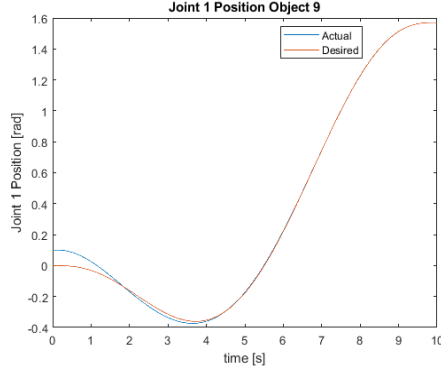


(c) Position Error with feedback

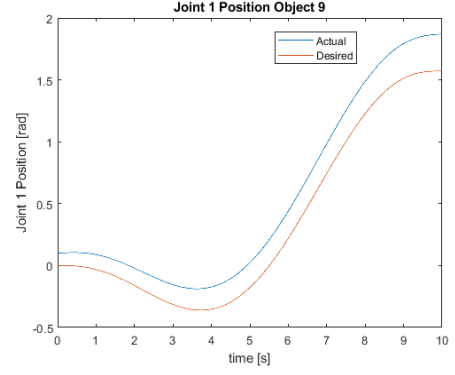


(d) Position Error without feedback

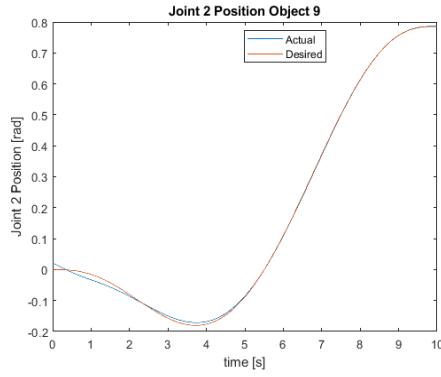
Figure 11.6: Position Errors on Object 7 and 8 With(left) and Without(right) feedback



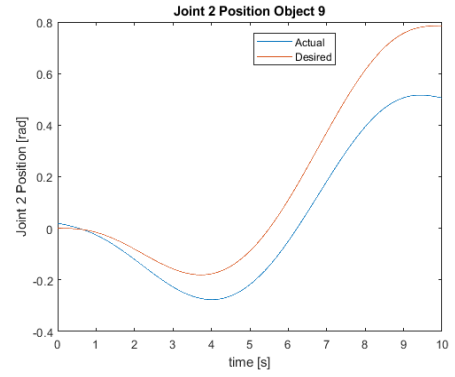
(a) Joint 1



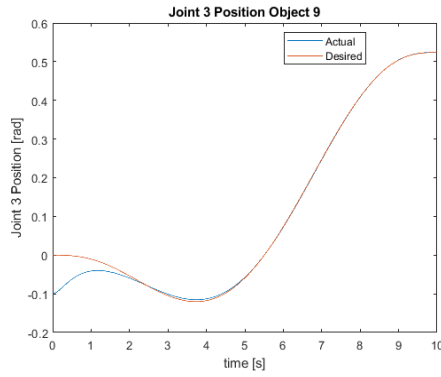
(b) Joint 1



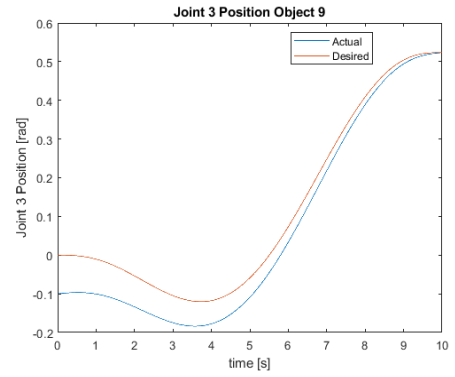
(c) Joint 2



(d) Joint 2



(e) Joint 3



(f) Joint 3

Figure 11.7: Tracking Performance on Object 9 with(left) and without(right) feedback.

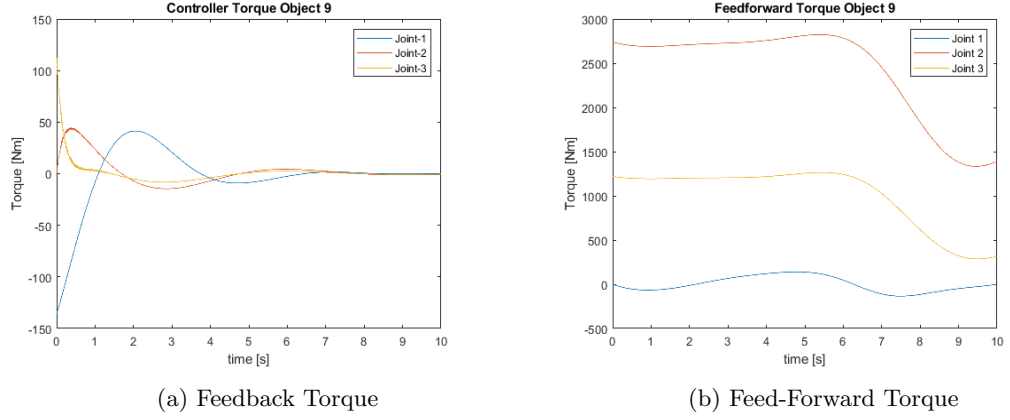


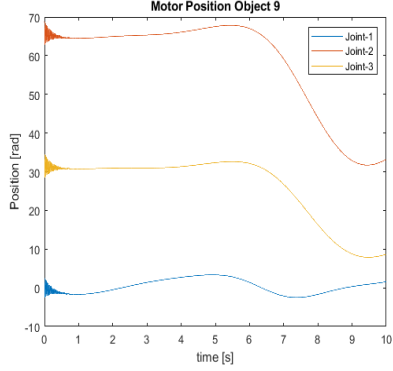
Figure 11.8: Feedback(left) and Feed-forward(right) Torque for object 9.

11.2.2 Feedback from Links and Motors Positions

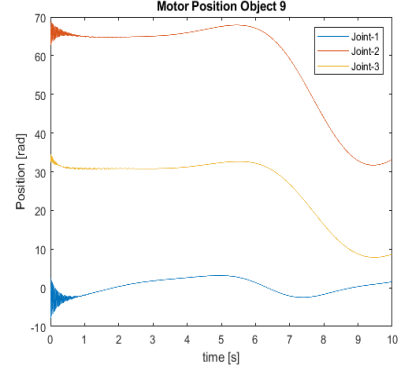
The control law in equation (11.2) helps us to guarantee the tracking of the desired trajectory with an error that decreases over time but we can observe a very oscillating transient phase in the motor response (see figure 11.9b) and for this reason, assuming that the motor position and velocity are measurable, we can update the previous control law with the one in equation (11.3)

$$\tau = \tau_A + K_p e_q + K_v \dot{e}_q + K_\theta e_\theta + K_{\dot{\theta}} \dot{e}_\theta, \quad (11.3)$$

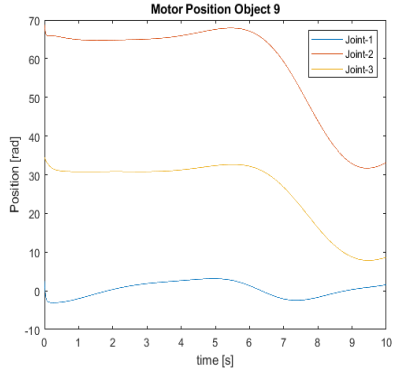
where $K_p > 0$, $K_v > 0$, $K_\theta > 0$ and $K_{\dot{\theta}} > 0$ for global asymptotic stability and which guarantees a better transient response of the motors as shown in figure 11.9c with $K_p = 1200$, $K_v = 800$, $K_\theta = 1.9$ and $K_{\dot{\theta}} = 1.5$.



(a) Motor position with control law in (10.1)



(b) Motor position with control law in (11.2)



(c) Motor position with control law in (11.3)

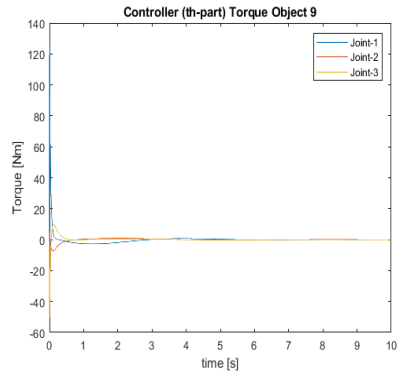
Figure 11.9: Motor position with object 9.

Figure (11.10) shows the contributions of the feedbacks due to the motors (figure 11.10a) and links (figure 11.10b) and their sum in figure 11.10c.

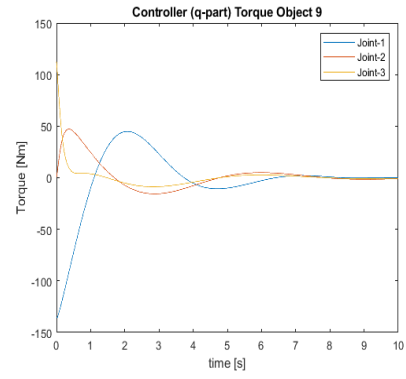
11.2.3 Remarks on the Numerical Regressor

Paying attention to the numerical regressor that we obtained using equation (3.4), we notice that there are some components with very small values. These components are associated to the inertia tensor of the object $I_o + m_o l_o^2$. From this observation, we try to compute the feed-forward torque assuming no information available on the object's inertia tensor but that only the mass and first moment of inertia are available. From equation (10.3) the object parameters vector will become

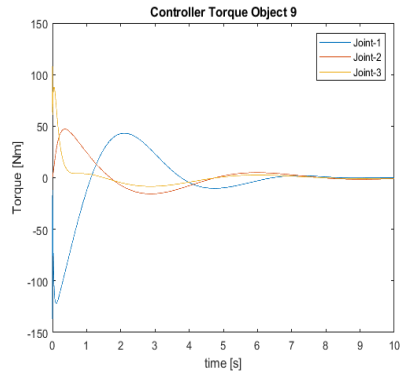
$$\pi_{\bar{o}} = [m_o \quad m_o l_{o_x} \quad 0 \quad 0 \quad 0]^T$$



(a) feedback torque due to motors



(b) feedback torque due to links



(c) feedback torque in (11.3)

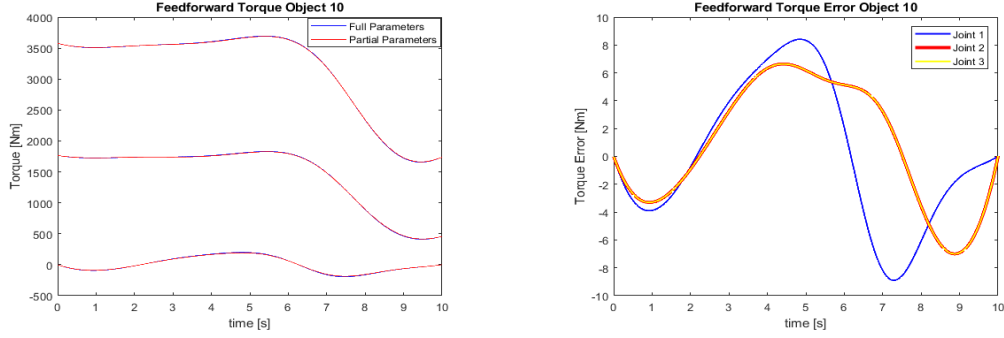
Figure 11.10: Feedback Torque in equation (11.3) with object 9.

This observation may be very useful in practice since we may have only partial information, mass and length of the object available to compute the feed-forward torque or we can also think about it as an uncertainty on the object parameters.

$$\tau = \bar{\tau}_A = \bar{Y}_o \bar{\pi}_o \quad (11.4)$$

The results of this assumption using object 10 on a SEA manipulator (can also be done for a rigid manipulator) are shown in figure 11.11 for the comparison of the feed-forward torque between complete and partial information on the object. Figure 11.12(right) shows the tracking performance when the obtained feed-forward torque is applied with accurate initial position and velocity of the links and motors and figure 11.12(left) when the control law in equation (11.5) is applied with inaccurate initial position and velocity of the links and motors.

$$\tau = \bar{\tau}_A + K_p e_q + K_v \dot{e}_q + K_\theta e_\theta + K_{\dot{\theta}} \dot{e}_\theta, \quad (11.5)$$

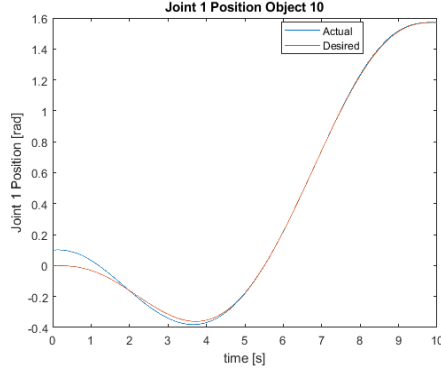


(a) Feed-Forward Torque with full and partial object information

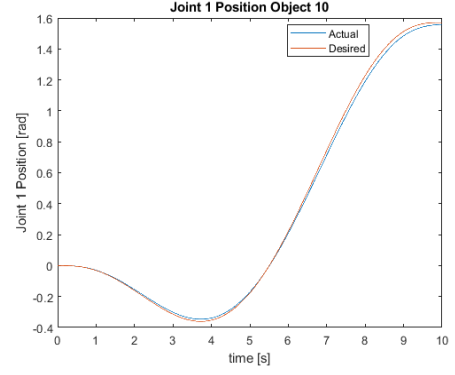
(b) Feed-Forward Torque Error with partial object information

Figure 11.11: Comparison of the Feed-forward torque obtained with full and partial information(left) and the Feed-forward torque Error obtained with partial information with respect to the one obtained with full information(right) for object 10.

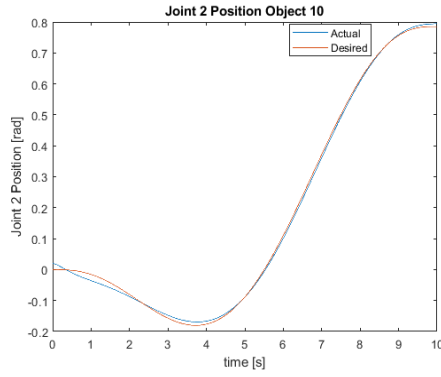
We can see from figure 11.12(left) how in presence of uncertainty in initial position and velocity or link and motor and also with partial or inaccurate information on the object, the control law in equation (11.5) is robust enough and still guarantees a good tracking of the desired trajectory with the same PD Gains used previously.



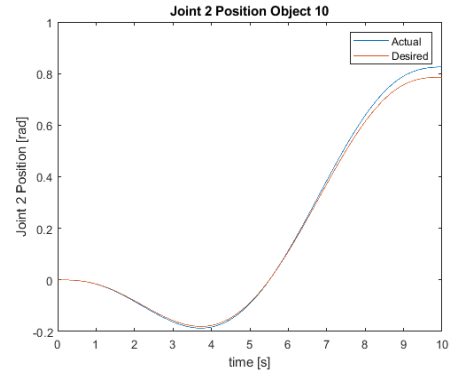
(a) Joint 1



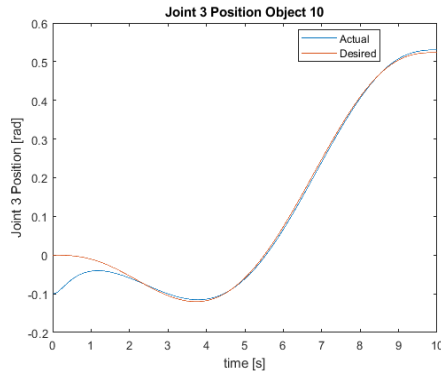
(b) Joint 1



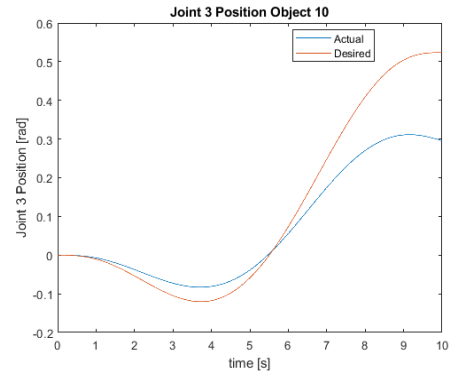
(c) Joint 2



(d) Joint 2



(e) Joint 3



(f) Joint 3

Figure 11.12: Tracking Performance with(left) and without(right) feedback with partial information on Object 10 .

Chapter 12

Conclusion

In this project, we have presented and evaluate in an experimental case study a new approach for the problem of tracking a desired trajectory, with different possible objects or payloads at the end-effector of a manipulator. This approach allows us to compute the feed-forward torque needed to track a desired trajectory thanks to a numerical regressor and some parameters of the object to be transported, avoiding to recompute the dynamics of the manipulator and this is its main advantage. To make it robust, we have added a feedback law that guarantees us a non-divergence and an attachment to the desired trajectory in the case of unknown initial conditions, inaccurate information about object parameters or any other uncertainties. We have seen how to proceed with a rigid and flexible manipulator with constant stiffness (SEA) and it could be further work to see if it can also be applied to variable stiffness robots (VSA) and also try to applied it on a real manipulator.

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